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REVIEW

From fundamentals to applications of incremental nonlinear dynamic inversion: A survey on INDI – Part I



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Abstract This paper, the first-part of a two part series of surveys on Incremental Nonlinear Dynamic Inversion (INDI), provides an overview of the evolution and developments in INDI. Written in a tutorial style, it presents different basic INDI variants and their specifics, such as model-based INDI, sensor-based INDI, and hybrid INDI. Furthermore, it sets these different approaches in context with each other. Later developments of INDI explicitly consider actuator dynamics. Those concepts are summarized and discussed in detail. Subsequently, studies that relate INDI to other control methods are summarized. Finally, an overview of various applications of INDI is given, covering different types of control loops and various types of vehicles and plants. This paper seeks to set these developments into context with each other. The purpose of this paper is twofold. INDI is already well-known in the domain of flight control but less so in other fields. Therefore, the paper is written in a comprehensive tutorial style to provide easy access to readers unfamiliar with the topic. On the other hand, the paper can serve as a reference for readers familiar with the topic.

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1. Introduction

Within the last decade, Incremental Nonlinear Dynamic Inversion (INDI) gained tremendous popularity. While in 2016, there were 186 publications on the topic, the number increased to 640 by 2020, and as of April 2024, there are 1530 publications, according to Google Scholar. Meanwhile, this topic has raised the interest of research groups all around the world.

While at the beginning INDI was mainly popular in the field of flight control, it started to draw the attention of control

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engineers in other domains in the recent years. Beyond flight control applications, like for instance multicopter,[1–4] transition drones,[5,6] fixed-wing aircraft,[7–10] and airships,[11,12] INDI has also been applied to an unmanned underwater vehicle,[13] spacecraft,[14,15] pressurized water reactor based small modular reactors,[16] to control a hydraulic hexapod flight simulator motion system,[17] and a pneumatic actuation system.[18]

The reason why INDI has become especially popular in flight control is because it addresses a major challenge of the already popular and widely used Nonlinear Dynamic Inversion (NDI). The key advantages of INDI over NDI are that it

(1) reduces the reliance on parts of the mathematical model of the system,
(2) improves disturbance rejection,
(3) is suitable for input non-affine systems.

While NDI requires precise model information to calculate control inputs, INDI uses measurements. In contrast to the model, the measurements contain information about the disturbances acting on the real system, allowing the INDI-based controller to reject the disturbances already one derivative level earlier than NDI. Furthermore, INDI is particularly suitable for novel aircraft configurations because model knowledge may be incomplete or inaccurate. Moreover, due to its ability to adequately address input non-affine systems in contrast to NDI, it can be used to control configurations with different flight domains such as X-35/ F-15, tilt-vehicles with tilting rotors or wings, and lift-to-cruise (e)VTOLs. The origins of NDI trace back to the 1960s,[19–21] and it has been successfully applied to many aircraft since Ref. 22–28, including the F-35.[29] Especially for classical fixed-wing aircraft, relatively accurate models can be obtained. On novel aircraft configurations, exact model knowledge might be lacking or too costly to identify, which is a challenge for NDI. Additionally, rapid development iterations with low-fidelity models are often desired and require control laws that still provide robust stability and performance. Here, INDI seems to be a solution, as it reduces a significant portion of the model knowledge-dependent part of NDI by appropriate measurements, as will be discussed in later sections.

INDI has the potential to provide robust performance, meaning adequate performance, even if the actual aircraft and the model used for control design have notable differences. However, INDI faces some challenges, discovered by several research groups independently in recent years, and various advancements to the baseline INDI have been developed.

As mentioned, INDI has gained impressive interest in flight control and has recently started attracting interest in other domains. While these fields are starting to explore the potential of INDI, we discovered the need for an overview that addresses the nuances and complexities of this approach. Through several projects at our institute in the past nine years and collaborations and discussions with other researchers in the field of INDI at conferences or joint projects, we have been fortunate to gain a comprehensive perspective on the various applications and developments of INDI.

This publication is the first part of a two-part survey series in INDI. The first part dives into the evolution of INDI. It provides an introduction and comparison of model-based, sensor-based, and hybrid INDI, as well as advanced variants such as E-INDI and ANDI, which both incorporate actuator dynamics. Additionally, this paper summarizes publications on how INDI familiarizes itself with other control methodologies and outlines its applications across various vehicle types and control loops.

The second part[30] of this two-part survey will focus on the modern trends in INDI and its components. It covers topics such as control allocation, specialized reference models, error controllers, synchronization techniques, and advances that have been made to these components in various publications in the INDI community. There, we also give an overview of different gain design approaches for INDI and research conducted regarding stability and robustness with respect to INDI.

Here, however, the first sections (Sections 2–7) focus on the evolution of different INDI variants and their characteristics. We will see that depending on the system characteristics (input affinity, relative degree, and actuator dynamics), different approaches to derive the incremental law exist and evolved over time. Furthermore, each approach provides unique insights and is valuable on its own; collectively, they contribute to a broader understanding.

To ease the readability and because our focus is to give the reader the idea of the concept, we will consider mostly scalar Single-Input-Single-Output (SISO) systems with a relative degree of one. Section 8 presents the INDI law for the more general case of Multi–Input–Multi–Output (MIMO) systems with a higher relative degree for completeness. Section 9 discusses extensions of INDI, which take into account actuator dynamics. After presenting the baseline INDI variants, Section 10 will give an outlook on additional components of INDI. As for each of those components several issues were raised and advances developed over time, we decided to summarize those variants and developments in the second part of this survey.[30] Section 11 addresses INDI and its relations to other control methods and related publications. Finally, Sections 12 and 13 give an overview of different applications of INDI. While Section 12 focuses on INDI and its application to different types of control loops, Section 13 gives an overview of the different vehicle and plant types INDI has been applied to.

2. Nonlinear dynamic inversion

Consider the following input affine SISO system

$$\begin{cases} \dot{\boldsymbol{x}} &= \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{G}(\boldsymbol{x}) \cdot u \\ y &= h(\boldsymbol{x}) \end{cases} \tag{1}$$

with states $\boldsymbol{x} \in \mathbb{R}^{n_x}$, control input u, controlled output y and sufficiently smooth mappings $\boldsymbol{f}, \boldsymbol{G}$ and h. The fundamental principle of NDI is to find a direct algebraic relation between the control input u and some time derivative $y^{(r)}$ of the controlled output y. This time derivative $y^{(r)}$ can be directly impacted by u, allowing shaping of the trajectory of y through appropriate choice of u. We hence denote $y^{(r)}$ as **responsive time derivative**. The system may then be described in a transformed way as

$$y^{(r)} = a_v(\boldsymbol{x}) + B_v(\boldsymbol{x}) \cdot u \tag{2}$$

calculated from consecutive differentiation of y, until the term $B_v(\boldsymbol{x})$ does not result in zero. The term r is denoted **relative degree**. Assuming that the transformed system dynamics are

exactly known, they can be inverted for u, resulting in the inversion law

$$u = B_v^{-1}(\boldsymbol{x}) \cdot \left(y^{(r)} - a_v(\boldsymbol{x})\right) \tag{3}$$

which can be used to calculate the u associated with a specific $y^{(r)}$. This inversion law can be used to calculate a required control input that needs to be commanded (u_{cmd}) to achieve a desired $y_{\text{des}}^{(r)}$. The inversion law is consequently expressed as

$$u_{\text{cmd}} = B_v^{-1}(\boldsymbol{x}) \cdot \left(y_{\text{des}}^{(r)} - a_v(\boldsymbol{x})\right) \tag{4}$$

In case of an exact inversion, all nonlinearities in the transformed system dynamics will cancel through the inversion law, leading to the resulting closed-loop input/output dynamics

$$y^{(r)} = y_{\text{des}}^{(r)} \tag{5}$$

with $y_{\text{des}}^{(r)}$ being the "new" (control) input. This new control input is denoted as **pseudo control** and is given the Greek symbol ν. Within the scope of this paper, we distinguish between the desired pseudo control $\nu_{\text{des}} = y_{\text{des}}^{(r)}$, and the "true" (plant response) pseudo control $\nu = y^{(r)}$. In other words, the pseudo control ν is equivalent to the responsive time derivative $y^{(r)}$. This notation is especially useful for MIMO systems, when multiple controlled outputs may have different relative degrees – the desired pseudo control will be denoted $\boldsymbol{\nu}_{\text{des}}$, and the true plant response will be denoted $\boldsymbol{\nu}$. Using the notation of pseudo control, the final SISO inversion law for NDI results as

$$u_{\text{cmd}} = B_v^{-1}(\boldsymbol{x}) \cdot (\nu_{\text{des}} - a_v(\boldsymbol{x})) \tag{6}$$

As mentioned before, NDI has been already applied successfully to several aircraft and other systems. For more details on NDI and its nuances the interested reader is referred to the following literature.[28,31,32] However, as can be directly seen from the inversion law, knowledge of the terms $a_v(\boldsymbol{x})$ and $B_v^{-1}(\boldsymbol{x})$ is crucial for the inversion to be successful. This makes the performance of NDI dependent on the quality of the mathematical model for the inversion.

3. Rise of sensor-based INDI

Around the beginning of this millennium, the first publications addressing the robustness issues of NDI regarding model knowledge arose.[23,26,33,34] Smith probably led the introduction of incremental laws using measurements of the pseudo control signals in 1998,[35] where he derives a "simplified nonlinear dynamic inversion" approach for the rotational control of an aircraft. The inversion law proposed by Smith structurally matches the later introduced INDI law, though Smith based his derivation directly on the rotational dynamics of an aircraft.

We want to showcase Smith's incremental law derivation to lead the reader to the topic. The derivation of Smith stands out from the other INDI derivations but results in the same incremental law and represents an intuitive view of incremental dynamic inversion. The derivation of Smith's inversion law will be showcased considering the single-state, SISO, and input affine system dynamics

$$\begin{cases} \dot{x} = f(x) + g(x) \cdot u \\ y = x \end{cases} \tag{7}$$

Furthermore, we assume a relative degree of one, resulting in the following pseudo control signal

$$\nu = \dot{y} = a_v(x) + B_v(x) \cdot u \tag{8}$$

where we define $a_v(x) = f(x)$ and $B_v(x) = g(x)$. Assuming that the pseudo control and control input are measurable (denoting the measurements in the following with subindex 0) and assuming that the sensor dynamics and actuator dynamics are linear, the following transfer functions are introduced

$$\begin{cases} \nu_0 = G_v(s) \cdot \nu \\ u_0 = G_u(s) \cdot u \\ u = G_{u_{\text{cmd}}}(s) \cdot u_{\text{cmd}} \end{cases} \tag{9}$$

where the respective transfer functions represent the dynamics of the respective sensor $G_v(s)$, filter $G_u(s)$, and actuator dynamics $G_{u_{\text{cmd}}}(s)$. Assuming furthermore a quasi-constant control effectiveness $B_v(x)$ (which means it changes slow as compared to the controlled dynamics), and interpreting the term $a_v(x)$ as separate disturbance signal allows to combine the linear transfer functions and the pseudo control dynamics from Eq. (8): First, the measured entities from Eq. (9) are introduced into Eq. (8) which leads, to

$$G_v^{-1}(s) \cdot \nu_0 = a_v(x) + B_v(x) \cdot G_u^{-1}(s) \cdot u_0 \tag{10}$$

Note that $G_v^{-1}(s)$ and $G_u^{-1}(s)$ may be improper transfer functions. Nevertheless, the mathematical derivation is carried on because simplifications at a later stage allow a proper realization. Second, the desired pseudo control is introduced by

$$\nu_{\text{des}} = a_v(x) + B_v(x) \cdot G_{u_{\text{cmd}}}(s) \cdot u_{\text{cmd}} \tag{11}$$

Then subtracting Eqs. (10), (11) from each other results in

$$\nu_{\text{des}} - G_v^{-1} \cdot \nu_0 = B_v \cdot \left(G_{u_{\text{cmd}}} \cdot u_{\text{cmd}} - G_u^{-1} \cdot u_0\right) \tag{12}$$

where (s) and (x) have been omitted for better readability. Note that the state-dependent term a_v cancels out at this stage. Finally, by isolating u_{cmd} on the left side, the following relation is obtained

$$\left(G_v \cdot B_v \cdot G_{u_{\text{cmd}}}\right) \cdot u_{\text{cmd}} = G_v \cdot \nu_{\text{des}} - \nu_0 + G_v \cdot B_v \cdot G_u^{-1} \cdot u_0 \tag{13}$$

which simplifies to

$$u_{\text{cmd}} = \frac{1}{G_u \cdot G_{u_{\text{cmd}}}} \cdot \left[u_0 + \frac{G_u}{G_v} \cdot B_v^{-1} \cdot (G_v \cdot \nu_{\text{des}} - \nu_0)\right] \tag{14}$$

If now it is assumed that the introduced transfer functions "are of a frequency content much greater than that of the required response bandwidth of the vehicle and have no saturating nonlinearities",[35] they may be omitted and replaced by a unity gain resulting in

$$u_{\text{cmd}} = u_0 + B_v^{-1} \cdot (\nu_{\text{des}} - \nu_0) \tag{15}$$

This corresponds to the INDI control law, as we will see in the following sections.

4. Sensor-based INDI - replacing the model dependent term by additional measurements

If the system is input affine, the analogy between NDI and INDI becomes clear. Therefore, we first introduce the NDI law and then replace a significant part of the model-dependent term by a relation depending on the measurements

of the output derivative $\dot{y}_0$ and measurements of the input u_0, in order to obtain the INDI law.

This derivation approach reveals clearly why INDI is often referred to as sensor-based approach and NDI in contrast as model-based approach. A similar way of deriving the INDI law, by rewriting the NDI law, can be found, e.g., in Ref. 36.

Given is the single state input affine system dynamics with a relative degree of one and invertible $g(x)$

$$\begin{cases} \dot{x} &= f(x) + g(x) \cdot u \\ y &= x \end{cases} \tag{16}$$

where x is the state, u is the control input, and y is the output. With pseudo control $v = \dot{x}$, the NDI law is obtained by directly solving Eq. (16) for u

$$u = g(x_0)^{-1} \cdot (v_{\text{des}} - f(x_0)) \tag{17}$$

where x_0 denotes the measurement of x. Assuming perfect inversion, this inversion law cancels all non-linearities, and for the closed-loop system consisting of plant and inversion law, the simple decoupled input–output relation

$$\dot{y} = v_{\text{des}} \tag{18}$$

is obtained. The relation given in Eq. (18) is also denoted as single-integrator form.[31]

The NDI law is denoted as an input transformation and decoupling inversion law because of the above-described properties. The inversion law Eq. (17) requires exact model knowledge in terms of the functions $f(x_0)$ and $g(x_0)$ and the contained parameters therein. The functions $f(x_0)$ and $g(x_0)$ are then calculated using the measurements x_0. However, if the function $f(x)$ is not known exactly, robustness issues can arise.[23,26,33,34]

Sensor-based INDI can address these robustness issues. Specifically, the dependency on exact model knowledge can be reduced by replacing $f(x_0)$ with the measurements of $\dot{y}_0$ and u_0. To obtain the necessary relationship, Eq. (16) is simply solved for $f(x)$:

$$f(x_0) = \dot{y}_0 - g(x_0) \cdot u_0 \tag{19}$$

Finally, the INDI law is obtained by replacing the not perfectly known $f(x_0)$ in the NDI law Eq. (17) by Eq. (19)

$$\begin{aligned} u &= g(x_0)^{-1} \cdot \Big[v_{\text{des}} - \underbrace{(\dot{y}_0 - g(x_0) \cdot u_0)}_{= f(x_0)} \Big] \\ &= g(x_0)^{-1} \cdot (v_{\text{des}} - \dot{y}_0) + u_0 \end{aligned} \tag{20}$$

where $\dot{y}_0$ and u_0 are measurements of the real system rather than mathematical models.

This shows very clearly why INDI is considered a sensor-based approach in contrast to NDI, which is model-based. It reveals how the state-only-dependent part of the model, given by the function $f(x)$, is replaced by additional sensor measurements. This reduction in dependency on exact model knowledge contributes to overcoming the major robustness issues from conventional NDI. The model dependency in the control effectiveness given by $g(x)$ remains. However, the required sensor signals might not always be trivial to obtain, and additional noise propagation might need to be treated with care.

5. Model-based INDI

The previous derivation showed why INDI is denoted as sensor-based in contrast to NDI, which is denoted as model-based. In the literature, we can also find a variant called "model-based INDI",[37] which may be interpreted as a combination of NDI and INDI, and which we want to present in the following. The following derivation builds conceptually on the concepts from references,[37–40] but the nomenclature and explanation have been adjusted to match the paper at hand.

Assumed is again an input affine, SISO, and scalar system Eq. (16) with relative degree one. The NDI law is given by equation Eq. (17), which requires model knowledge $g(x_0)$ and $f(x_0)$. Similar to the sensor-based INDI derivation in Section 4, we explicitly replace $f(x_0)$ in the NDI law according to equation Eq. (19). The subtle difference here is that we utilize a model-based estimation, provided by an On-Board Model (OBM)

$$f_{\text{OBM}}(x_0) = \dot{y}_{\text{OBM}} - g(x_0) \cdot u_{\text{OBM}} \tag{21}$$

instead of $f(x_0)$. This leads to the model-based INDI law

$$\begin{aligned} u &= g(x_0)^{-1} \cdot \Big[v_{\text{des}} - \underbrace{(\dot{y}_{\text{OBM}} - g(x_0) \cdot u_{\text{OBM}})}_{= f_{\text{OBM}}(x_0)} \Big] \\ &= g(x_0)^{-1} \cdot (v_{\text{des}} - \dot{y}_{\text{OBM}}) + u_{\text{OBM}} \end{aligned} \tag{22}$$

where $\dot{y}_{\text{OBM}}$ and u_{OBM} are calculated from mathematical models rather than being measurements of the real system (as would be the case for sensor-based INDI).

The inversion law Eq. (22) can also explicitly be written in an "incremental form", hence commanding an additional increment on the current control input estimation, which results in

$$u = u_0 + \Delta u = u_0 + g(x_0)^{-1} \cdot (v_{\text{des}} - \dot{y}_{\text{OBM}}) \tag{23}$$

The origins of model-based INDI may be attributed to Bordignon et al., who describe that structure in their 2002 paper on controlling the X-35B aircraft.[41] Its successor aircraft, the F-35, is described to use the same control concept.[42] The model-based INDI used in Ref. 42 is depicted in Fig. 1.

Note 1. As Fig. 1 shows, there is no explicit model of an actuator. In fact, an increment Δu (denoted as δ_{cmd}), which is added to the previous time step's control command u_{cmd}, provides the control input command. For actuators that are fast and accurate enough, this may hold. In that case, one assumes that the actuator achieves the commanded increment within one time step.

Note 2. Although they are very similar, a subtle difference exists between NDI and Model-Based INDI, which is in Section 7, Note 6, will clarify after the necessary additional derivations are explained.

6. Hybrid INDI

In 2018, Ref. 42 reported that in flight domains with high nonlinearities, in which high model errors occur, e.g., at a high angle of attack, the model-based approach deteriorates. In order to accommodate for modeling errors and improve the robustness, they introduce the augmentation depicted in Fig. 2. The suggested augmentation uses approximations of the model error based on measured parameters to calculate an additional acceleration error signal. Approaches such as

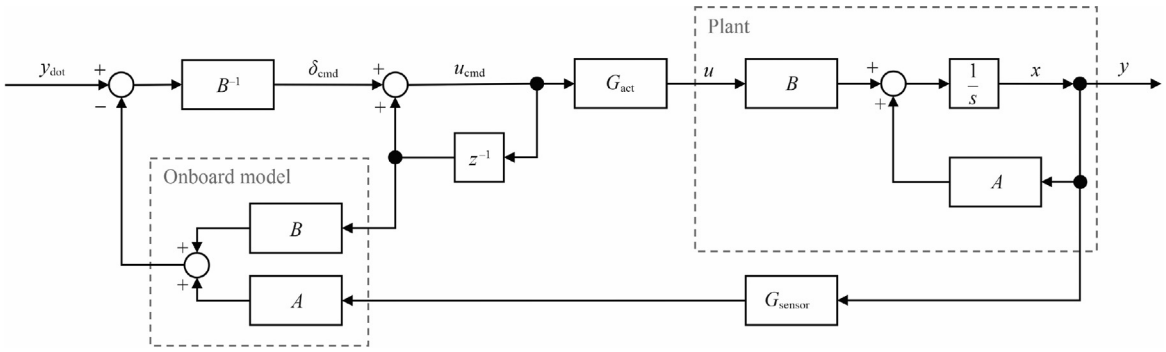


Fig. 1 Model-based INDI of F-35 Joint Strike Fighter adapted from Ref. 42.

that of Ref. 42, which use a combination of model-based and sensor-based INDI, will be referred to as Hybrid INDI in the following. We will present two categories of Hybrid INDI: (A) based on a proportional blend, Section 6.1, (B) based on a frequency-dependent blend, Section 6.2.

6.1. Proportional blend

In 2020, Ref. 38 combined sensor-based and model-based INDI to what they later call Hybrid INDI in 2021.[37] The approach is based on the idea to augment the model-based INDI by proportional feedback of the error between the measurement $\dot{y}_0$ and the model-based estimate $\dot{y}_{\mathrm{OBM}}$ as depicted in Fig. 3.

In the following we will demonstrate that the resulting control law corresponds to a linear proportional blend between the model-based and sensor-based output derivatives. We adjusted the nomenclature in the following derivation compared to Fig. 3, as follows: $\dot{y}_0 = \dot{x}_0 = \dot{x}_{\mathrm{meas}}, \dot{y}_{\mathrm{OBM}} = \dot{x}_{\mathrm{obm}}, \dot{y}_{\mathrm{add}} = \dot{x}_{\mathrm{add}}, \Delta d = \Delta \nu$ and $\nu_{\mathrm{des}} = \dot{x}_{\mathrm{des}}$. The augmentation

$$\dot{y}_{\mathrm{add}} = K_{\mathrm{aug}} \cdot (\dot{y}_0 - \dot{y}_{\mathrm{OBM}}) \tag{24}$$

is subtracted from the pseudo control increment $\Delta\nu = \nu_{\mathrm{des}} - \dot{y}_{\mathrm{OBM}}$ such that the model-based INDI control law given in Eq. (23) becomes

$$u = u_0 + g(x_0)^{-1} \cdot \left[\nu_{\mathrm{des}} - \dot{y}_{\mathrm{OBM}} - K_{\mathrm{aug}} \cdot (\dot{y}_0 - \dot{y}_{\mathrm{OBM}})\right] \tag{25}$$

where the term

$$\begin{aligned} &\dot{y}_{\mathrm{OBM}} + K_{\mathrm{aug}} \cdot (\dot{y}_0 - \dot{y}_{\mathrm{OBM}}) \\ &\quad = (1 - K_{\mathrm{aug}}) \cdot \dot{y}_{\mathrm{OBM}} + K_{\mathrm{aug}} \cdot \dot{y}_0 = \dot{\hat{y}} \end{aligned} \tag{26}$$

can be interpreted as an estimate of $\dot{y}$, given as a proportional linear blend between the sensor-based (measured) output derivative $\dot{y}_0$ and the model-based output derivative $\dot{y}_{\mathrm{OBM}}$.

The factor K_{aug} may depend on system states and, therefore, could enable a flight condition-dependent weighting between the model-based estimate and the measurement. In flight regimes where the model is known accurately, relying more on the model makes sense as a feedback reduction increases the stability margins, as shown in Ref. 38. However, in conditions like high angle of attack (AoA), where the model may be more inaccurate, one may rely more on the measurements.

6.2. Frequency-dependent blend

Instead of using a linear proportional blend between $\dot{y}_0$ and $\dot{y}_{\mathrm{OBM}}$, which fixes their ratio over the complete frequency range, a complementary filter can be used in order to achieve a frequency-dependent weighting, as Ref. 2 proposed in 2018.

The derivative $\dot{y}_0$ in the INDI law Eq. (20) is then replaced by the estimate $\dot{\hat{y}}$. This estimate is calculated as follows

$$\dot{\hat{y}}(s) = (1 - L(s)) \cdot \dot{y}_{\mathrm{OBM}}(s) + L(s) \cdot \dot{y}_0(s) \tag{27}$$

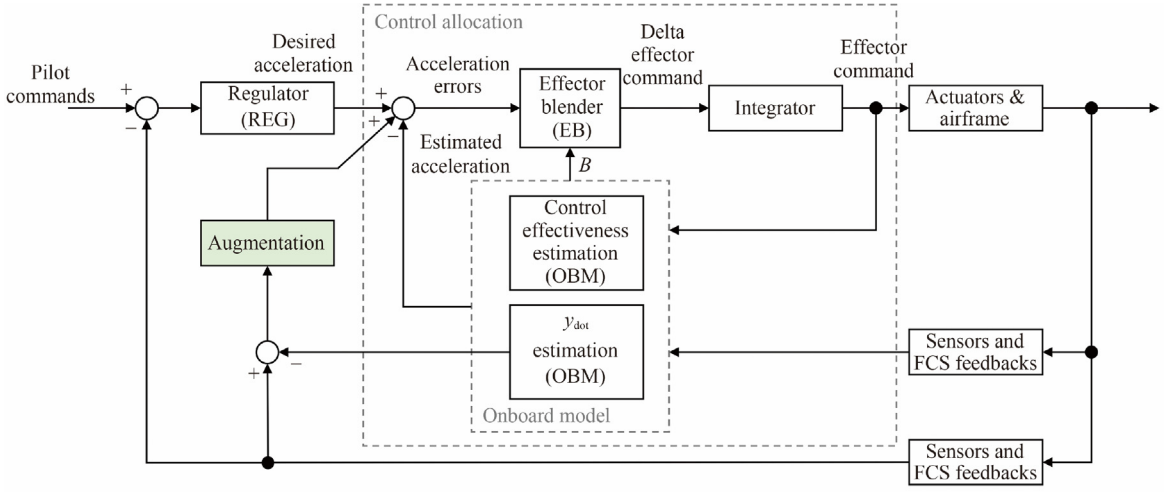


Fig. 2 Augmented NDI control law adapted from Ref. 42.

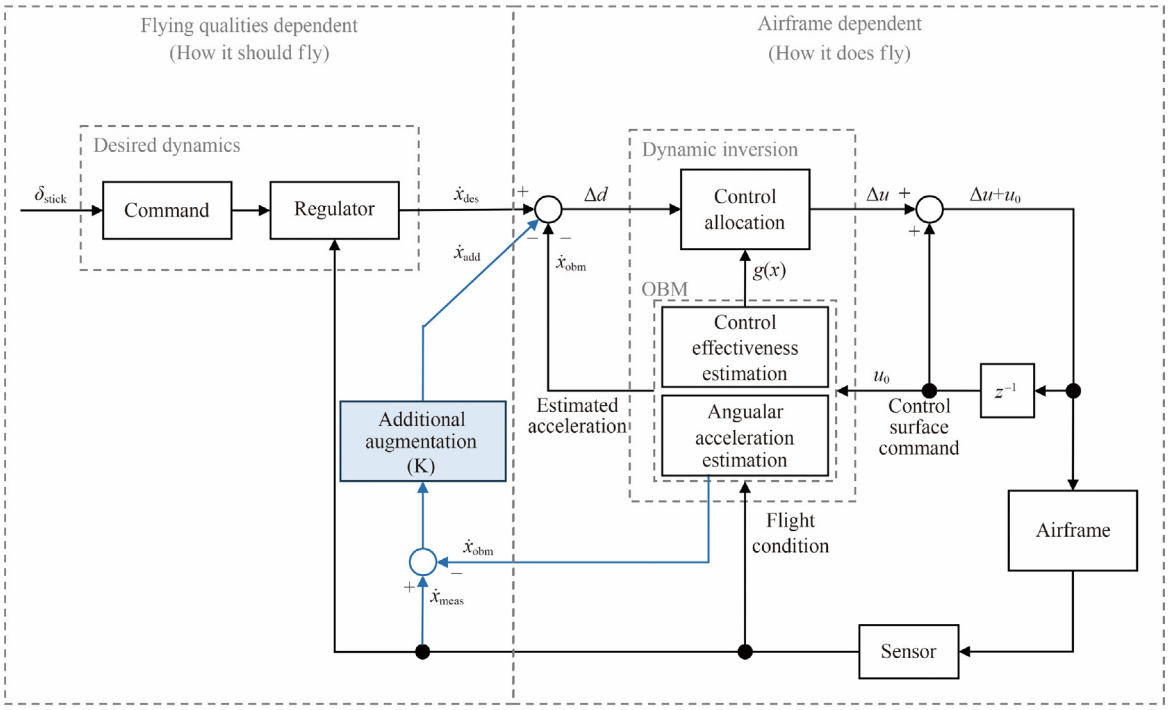


Fig. 3 Hybrid INDI adapted from Ref. 38.

where $L(s)$ is a linear first-order low-pass filter. Consequently, $1 - L(s)$ becomes a high pass filter, and $\dot{y}_0$ represents the measured translational accelerations. For $\dot{y} = \dot{\omega}$, being the angular accelerations, Akkinapalli uses linear second-order lowpass filters and calculates the measured derivatives via $\dot{y}_0 = s \cdot y_0$. For this case Eq. (27) results in

$$\hat{\dot{y}} = \frac{s^2 + 2\zeta\omega_0 \cdot s}{s^2 + 2\zeta\omega_0 \cdot s + \omega_0^2} \cdot \dot{y}_{\mathrm{OBM}} + \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 \cdot s + \omega_0^2} \cdot s \cdot y_0 \tag{28}$$

With the complementary filter approach, only the low-frequency content of the sensor-based signal $\dot{y}_0$ is used, and therefore, high-frequency noise, higher order dynamics, structural mode pick up, and vibrations are eliminated or attenuated. These effects then do not propagate to the control input, where they could probably cause oscillations, amplify control noise, contribute to accelerated wear out, or excite structural modes.

The model-based estimate of the derivative $\dot{y}_{\mathrm{OBM}}$ provides the high-frequency part in the final estimate used in the INDI control law. Hence, steady-state errors due to model uncertainties have no influence. Furthermore, the high-frequency part of the model-based estimate is usually relatively accurate. Ref. 43 recommended the same approach as Akkinapalli and discuss the initialization of the complementary filter. Furthermore, they extend the complementary filter to account for sensor dynamics and delays and incorporate additional filtering, such as roll-off or notch filters, into the complementary filter design. Finally, they compare analytically and in simulations several approaches to obtain $\hat{\dot{y}}_0$, and the related synchronization techniques. The importance of synchronization and the different approaches proposed in the literature to estimate $\dot{y}_0$ are addressed in more detail in the second part of this survey in Ref. 30.

In 2022, Ref. 40 introduced another approach that is also dubbed Hybrid INDI (not to be confused with the Hybrid INDI from Ref. 38). It uses a different complementary filter variant with a suitable synchronization scheme on the actuator feedback path. The filter is given by

$$\hat{\dot{y}} = \frac{s^2}{s^2 + K_P \cdot s + K_I} \cdot \dot{y}_{\mathrm{OBM}} + \frac{K_P \cdot s + K_I}{s^2 + K_P \cdot s + K_I} \cdot s \cdot y_0 \tag{29}$$

Choosing $K_P = 2\zeta\omega_0$ and $K_I = \omega_0^2$ results in

$$\hat{\dot{y}} = \frac{s^2}{s^2 + 2\zeta\omega_0 \cdot s + \omega_0^2} \cdot \dot{y}_{\mathrm{OBM}} + \frac{2\zeta\omega_0 \cdot s + \omega_0^2}{s^2 + 2\zeta\omega_0 \cdot s + \omega_0^2} \cdot s \cdot y_0 \tag{30}$$

and shows that this approach differs from Eq. (28) in the choice of the numerator of the model and measurement part.

In 2023, Ref. 44 proposed an extended state observer (ESO) to estimate angular accelerations based on an onboard mathematical model and rate measurements, which can be shown to equal Eq. (28). Ref. 44 showed empirically and by frequency domain analysis that ESO and hence the complementary filter in Eq. (28) can achieve improved noise attenuation compared to the complementary filter in Eq. (29). We elaborate on that in more detail in the second part of this survey in Ref. 30.

The hybrid INDI with complementary filter can be viewed as INDI with a pseudo control estimate ν calculated by a complementary filter. The measured pseudo control could be also estimated using other approaches as for example Kalman filter based methods. Examples for Kalman filter based approaches in combination with INDI are provided by Ref. 45–47.

7. Derivation for input non-affine systems - Taylor's approximation

Around the same time when Smith flight tested his incremental inversion law[35] on the VAAC Harrier,[10] Bacon and Ostroff[33] published an extension to Smith's approach. Their derivation of the incremental inversion law is more generic, with the main advantage of also coping with input non-affine control inputs

and is based on Taylor's approximation. It is the nowadays most common approach to derive the INDI control law.[3,11,12,40,48–55]

In the following, we derive Bacon and Ostroff's inversion law from 2000[33] with adjusted nomenclature for an input non-affine single-state SISO system

$$\begin{cases} \dot{x} &= F(x,u) = f(x) + g(x,u) \\ y &= h(x) \end{cases} \tag{31}$$

with a relative degree of one such that

$$\dot{y} = \frac{\partial h(x)}{\partial x} \cdot \dot{x} = \frac{\partial h(x)}{\partial x} \cdot (f(x) + g(x,u)) \tag{32}$$

with $\frac{\partial h(x)}{\partial x} \cdot g(x,u)$ being non- zero. Here, $g(x,u)$ can be a nonlinear function of u and of such a form that we cannot solve Eq. (32) anymore directly for the control input u as we did in the previous derivations.

Bacon and Ostroff propose to form the first-order Taylor approximation around x_0 and u_0 as follows

$$\begin{aligned} \dot{x} = F(x,u) \quad &\approx F(x_0,u_0) \\ &+ \frac{\partial F(x,u)}{\partial x}\bigg|_{x=x_0,u=u_0} \cdot (x - x_0) \\ &+ \frac{\partial g(x,u)}{\partial u}\bigg|_{x=x_0,u=u_0} \cdot (u - u_0) \end{aligned} \tag{33}$$

where $x_0 = x(t - T_s)$ and $u_0 = u(t - T_s)$ are the state and input from the recent past, that is, delayed by sampling time T_s. Substituting $F(x_0,u_0) = \dot{x}_0, \Delta u = u - u_0$

$$\frac{\partial F(x,u)}{\partial x}\bigg|_{x=x_0,u=u_0} = A_x \tag{34}$$

and

$$\frac{\partial g(x,u)}{\partial u}\bigg|_{x=x_0,u=u_0} = B_x \tag{35}$$

results in

$$\dot{x} \approx \dot{x}_0 + A_x \cdot (x - x_0) + B_x \cdot \Delta u \tag{36}$$

A_x represents the current sensitivity of the state derivative with respect to the state. Note that this would be equivalent to the system matrix $\boldsymbol{A}$, resulting from linearizing the system around x_0 and u_0 corresponding to trim condition or operating point as a first-order Taylor approximation is indeed a linearization.

B_x denotes the current sensitivity of the state derivatives with respect to the control input u and hence corresponds to the linearized input, or $\boldsymbol{B}$ matrix at the current state and input position (and not trim condition). The first derivative y of the outputs is given by

$$\dot{y} = \frac{\partial h(x)}{\partial x} \cdot \dot{x} \tag{37}$$

and by inserting the first-order Taylor approximation of $\dot{x}$ around x_0 and u_0, Eq. (36), we obtain

$$\dot{y} \approx \frac{\partial h(x)}{\partial x} \cdot [\dot{x}_0 + A_x \cdot (x - x_0) + B_x \cdot \Delta u] \tag{38}$$

Assuming a relative degree of one for the output y, then the first derivative of the output $\dot{y}$ is impacted by the control input u, and therefore, the first output derivative represents the pseudo control: $v = \dot{y}$. We can reformulate Eq. (38) accordingly to

$$v \approx \dot{y}_0 + A_v \cdot (x - x_0) + B_v \cdot \Delta u \tag{39}$$

where $\dot{y}_0 = \frac{\partial h(x)}{\partial x} \cdot \dot{x}_0, A_v = \frac{\partial h(x)}{\partial x} \cdot A_x$ and $B_v = \frac{\partial h(x)}{\partial x} \cdot B_x$. The inversion law for Δu will provide the required change in the control input necessary to achieve a desired change in pseudo control $\Delta v = v_{\text{des}} - \dot{y}_0$

$$B_v \cdot \Delta u = (v_{\text{des}} - \dot{y}_0) - A_v \cdot (x - x_0) \tag{40}$$

Assuming "a sufficient control update rate, x approaches x_0",[33] such that the term $A_v \cdot (x - x_0)$ can be neglected. For overactuated systems, one can solve the expression for Δu using various Control Allocation methods or inverses.

Note that Ref. 33 is referring to measuring $\dot{x}_0$ and calculating $\dot{y}_0$ online, which requires the calculation and modeling of $h_x = \frac{\partial h(x)}{\partial x}$. Fig. 4[33] depicts their inversion law, which was further investigated in Ref. 56. Note that in Fig. 4, δ denotes the control input u, and the control allocation is performed using a weighted pseudo inverse.

In 2010, Ref. 49 revisited Bacon's[33,56] inversion laws. They introduce the name "Incremental Nonlinear Dynamic Inversion" and its abbreviation INDI. They derive the INDI law directly for the rotational dynamics of an aircraft using Taylor's approximation. Thereby, they make the following important assumptions.[49]

Initially, they assume that the sensors, which measure the angular accelerations, are ideal. However, as they point out, this assumption does not hold for real systems, and a delay and/or deterioration of the angular acceleration measurement/estimation must be considered. Later, they introduce a linear predictive filter to estimate the angular accelerations. They also find that the ideal INDI inversion law is sensitive to time delays in the angular acceleration feedback. They specifically mention the difficulty in obtaining angular acceleration estimations required for the INDI inversion laws.

Furthermore, they consider the control effectors to be ideal; that is, the commanded control surface deflection is achieved instantaneously (infinitely fast actuator dynamics). In such a case, the control surface instantly produces the desired angular acceleration increment, and a difference in the state does not have time to build up, as it is one integration above. Hence, the state-dependent terms $A_x(x - x_0)$ in Eq. (36) can be neglected. Later, this assumption was confirmed by Ref. 57 in 2023. There, the authors use NDI, taking first-order actuators into account. A suitable formulation of the desired error dynamics shows that for a high actuator bandwidth, the state-dependent terms become negligible, and the NDI law results in the INDI law. Section 9 will address this in more detail.

Note 3. It is worth noting that, indeed, if we apply Bacon and Ostroff's concept to the input affine system that we used in the previous sections given by Eq. (16), the resulting inversion law will correspond to the sensor-based INDI law Eq. (20). The Taylor approximation of system Eq. (16) results in

$$\begin{aligned} \dot{x} \approx \dot{x}_0 \quad &+ \frac{\partial f(x)}{\partial x}\bigg|_{x=x_0} \cdot \Delta x + \frac{\partial (g(x)\cdot u)}{\partial x}\bigg|_{x=x_0,u=u_0} \cdot \Delta x \\ &+ g(x) \cdot \Delta u \end{aligned} \tag{41}$$

Neglecting again the terms including Δx, and using that the output of system Eq. (16) is $y = x$ and the pseudo control variable is $v = \dot{y}$, Eq. (41) is solved for u

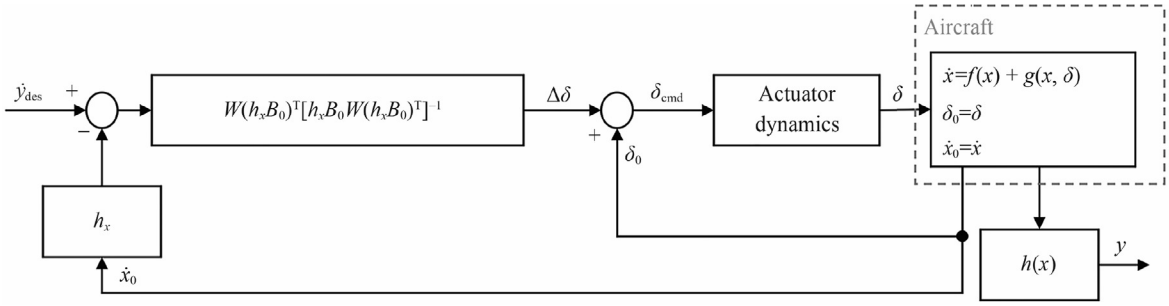


Fig. 4 INDI controller adapted from Ref. 33.

$$u = g(x_0)^{-1} \cdot (\nu_{des} - \dot{y}_0) + u_0 \tag{42}$$

where $\nu = \dot{y} = \dot{x}$ and $\Delta u = u - u_0$ is used. We see that this resulting INDI law corresponds to the sensor-based INDI law Eq. (20), derived in the previous sections based on NDI.

The sensor-based INDI law seems not to require any assumptions or approximations, whereas the just introduced Taylor-based derivation uses $\Delta x = 0$. It is a justified question to ask: why does the same inversion law result? The answer is: No error was introduced through the Taylor-based law for the considered system, either - the approximations and assumptions did not come to bear.

This is briefly explained in the following: Considered is a SISO, scalar, and input affine system with relative degree one and invertible $g(x)$, as given in Eq. (16). Assuming its existence, a Taylor series expansion around x_0 and u_0 can be written as

$$\begin{aligned} \dot{y} = \dot{y}_0 \quad & + \frac{\partial(f(x)+g(x)\cdot u)}{\partial x}\bigg|_{x=x_0,u=u_0} \cdot \Delta x + x_{\text{h.o.t.}} \\ & + \frac{\partial(f(x)+g(x)\cdot u)}{\partial u}\bigg|_{x=x_0,u=u_0} \cdot \Delta u + u_{\text{h.o.t}} \end{aligned} \tag{43}$$

where $\Delta x = x - x_0, \Delta u = u - u_0$ and where $x_{\text{h.o.t.}}$ are all higher order terms in x, and $u_{\text{h.o.t.}}$ all higher order terms in u. Eq. (43) represents a Taylor series expansion and not a Taylor approximation; hence, the equal sign "=" is valid. Now, for the system under consideration, the following two simplifications can be made:

Input affinity: We consider input affine systems - that means they are linear in the control input u. Therefore $\partial(f(x) + g(x) \cdot u)/\partial u = g(x)$, and all higher order terms result in 0. In other words, without approximation, $u_{\text{h.o.t}} = 0$.

Evaluation at x_0: Similar to the derivation via direct inversion as presented in Section 4, we evaluate the Taylor series expansion "function" Eq. (43) at $x = x_0$ such that $\Delta x = x - x_0 = 0$. Since a desired $\dot{y}_{des}$ may be achieved by altering the state x, or by altering the control input u, or both, in theory, there are infinitely many possible combinations. However, we cannot instantaneously change the state, but we can change the control input! For infinitely fast control inputs, choosing $x = x_0$ is justified and possibly the only reasonable choice. In that case, also all higher order terms, and thus $x_{\text{h.o.t}}$, result in 0.

Hence, for input affine systems, evaluated at the current state $x = x_0$, holds that

$$\dot{y} = \dot{y}_0 + g(x) \cdot \Delta u \tag{44}$$

Summarized, the input affinity and the choice $\Delta x = x - x_0 = 0$ causes the Taylor approximation to not be an approximation but be equivalent to the "full" Taylor series expansion. This results in the same inversion law as is obtained by the sensor-based derivation in Section 4.

Note 4. With the Taylor approximation-based approach, an INDI law can also be obtained for nonlinear systems which are input non-affine and where $g(x, u)$ might be a nonlinear function in u. This is, for example, the case for VTOLs with tilting wings or rotors.[58]

Then, the Taylor-based approach linearizes the control effectiveness $g(x, u)$ locally around the current state x_0 and input u_0, allowing to form the INDI law for sufficiently small Δu. The fact that INDI can be successfully applied to input non-affine systems was, for example, demonstrated in Refs. 5,59.

Note 5. Another approach for input non-affine systems is denoted as sensory NDI by Milz et al.[58,60]. Consider the following input non-affine system

$$\begin{cases} \dot{\boldsymbol{x}} &= \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}) \\ \boldsymbol{y} &= \boldsymbol{x} \end{cases} \tag{45}$$

For simplicity, we assume all outputs to have a relative degree of one. The desired dynamics are given by

$$\boldsymbol{\nu}_{des} = \boldsymbol{f}(\boldsymbol{x}_0) + \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_{cmd}) \tag{46}$$

where $\boldsymbol{x}_0$ denotes the measured states. They are solved for $\boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_{cmd})$, resulting in

$$\boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}) = \boldsymbol{\nu}_{des} - \boldsymbol{f}(\boldsymbol{x}_0) = (\boldsymbol{\nu}_{des} - \dot{\boldsymbol{y}}_0) + \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_0) \tag{47}$$

Equivalent to the derivations in Section 4, $\boldsymbol{f}(\boldsymbol{x}_0)$ can be replaced by $\dot{\boldsymbol{y}}_0 - \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_0)$ where $\dot{\boldsymbol{y}}_0$ and $\boldsymbol{u}_0$ denote the measurements of $\dot{\boldsymbol{y}}$ and $\boldsymbol{u}$. Hence, the name sensory NDI.

Now, the input non-affine control allocation problem Eq. (47) can be approximated in the neighbourhood of $\boldsymbol{u}_T$ through a first-order Taylor series approximation of $\boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_{cmd})$ given by

$$\boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}) \approx \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_T) + \frac{\partial \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u})}{\partial \boldsymbol{u}}\bigg|_{\boldsymbol{u}=\boldsymbol{u}_T} \cdot (\boldsymbol{u} - \boldsymbol{u}_T) \tag{48}$$

where $\boldsymbol{u}_T$ is a selected linearization point of the Taylor series approximation used in the INDI controller. If we insert Eq. (48) into Eq. (47), the implicit inversion law Eq. (47) reduces

to the input affine sensory NDI law and we can solve for $\boldsymbol{u}$. Then we obtain

$$\left.\frac{\partial \boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_T}\cdot \boldsymbol{u} = (\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_0)+\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u}_0)-\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u}_T)+\left.\frac{\partial \boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_T}\cdot \boldsymbol{u}_T \tag{49}$$

If the Taylor approximation is performed at the current measured input $\boldsymbol{u}_T=\boldsymbol{u}_0$ then $\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u}_0)$ and $\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u}_T)$ cancel and we obtain

$$\left.\frac{\partial \boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_0}\cdot \boldsymbol{u} = (\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_0)+\left.\frac{\partial \boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_0}\cdot \boldsymbol{u}_0 \tag{50}$$

Alternatively, for MIMO systems with overactuation, we obtain the control allocation problem

$$\boldsymbol{u}=\boldsymbol{B}_v^{-1}\cdot(\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_0)+\boldsymbol{u}_0 \tag{51}$$

Alternatively, for MIMO systems with overactuation, we obtain the control allocation problem

$$\boldsymbol{B}_v\cdot\boldsymbol{u}=\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_0+\boldsymbol{B}_v\cdot\boldsymbol{u}_0 \tag{52}$$

Here, in the case of an overactuated MIMO system, we can notice a difference to the previously derived INDI laws, while for SISO systems or MIMO systems with equal number of inputs and outputs, indeed, the resulting inversion law Eq. (51) corresponds to the conventional derived INDI laws. So far, all INDI laws had the incremental form $\boldsymbol{u}_{\text{cmd}}=\boldsymbol{u}_0+\Delta\boldsymbol{u}$, where the control allocation provided the $\Delta\boldsymbol{u}$ (Eqs. (15), (20), (23), (40)). That means all INDI laws so far are used to calculate an increment command $\Delta\boldsymbol{u}$. For the sensory NDI MIMO law Eq. (52), however, one explicitly calculates the term $\boldsymbol{B}_v\boldsymbol{u}_0$ and then solves the control allocation problem for a full $\boldsymbol{u}_{\text{cmd}}$. In summary, for overactuated systems or nonlinear control allocation, this will lead to different results compared to the incremental allocation. However, for both approaches, the incremental allocation provides a solution that is only valid in the vicinity of the linearization point $\boldsymbol{u}=\boldsymbol{u}_T$. Further, it is limited in accuracy, as a linear approach approximates the nonlinear problem. In such cases virtual controls might substitute control variables to increase accuracy. For example, Ref. 58 and Ref. 1 substituted the propeller speed as control variable $u=\omega_p$ by ω_p^2 since the influence of the propeller speed on the thrust is mainly quadratic.

Note 6. In the following we address the differences between NDI and model-based INDI. In Section 4, the NDI law for the system Eq. (16) is given by Eq. (17), which is repeated here for convenience

$$u=g(x_0)^{-1}\cdot(v_{\text{des}}-f(x_0)) \tag{53}$$

On the other hand, the model-based INDI law for the same system Eq. (16) follows according to Eq. (22) as

$$u=g(x_0)^{-1}\cdot\Big[v_{\text{des}}-\underbrace{(f(x_0)+g(x_0)\cdot u_0)}_{=\dot{y}_{\text{OBM}}}\Big]+u_0 \tag{54}$$

which are very similar, except for the intentional adding and subtracting of the term $g(x_0)u_0$.

The added $g(x_0)u_0$ together with $f(x_0)$ forms a model-based estimate of the pseudo control $\dot{y}_{\text{OBM}}$. The subtracted part $g(x_0)u_0$can be brought out of the inverse to enable the allocation of an increment $\Delta u=u-u_0$. Both NDI and model-based INDI require the mathematical models represented as $f(x_0)$ and $g(x_0)$. However, model-based INDI furthermore requires estimating the current control input u_0. Furthermore, the difference between NDI and model-based INDI is system-dependent, which we will illustrate in the following four cases:

Case 1. MIMO: $n_u=n_y$

Input affine: In case (the matrix) $\boldsymbol{g}(\boldsymbol{x}_0)$ is invertible, NDI and model-based INDI are equivalent, since $\boldsymbol{g}(\boldsymbol{x}_0)^{-1}\cdot\boldsymbol{g}(\boldsymbol{x}_0)=\boldsymbol{I}$, where $\boldsymbol{I}$ is the identity matrix.

Case 2. MIMO: $n_u>n_y$

Input affine: Here for NDI, the control allocation problem

$$\boldsymbol{g}(\boldsymbol{x}_0)\cdot\boldsymbol{u}=\boldsymbol{v}_{\text{des}}-\boldsymbol{f}(\boldsymbol{x}_0) \tag{55}$$

needs to be solved, and for the model-based INDI the control allocation problem

$$\boldsymbol{g}(\boldsymbol{x}_0)\cdot\boldsymbol{u}=\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_{\text{OBM}}+\boldsymbol{g}(\boldsymbol{x}_0)\cdot\boldsymbol{u}_0 \tag{56}$$

with $\dot{\boldsymbol{y}}_{\text{OBM}}=\boldsymbol{f}(\boldsymbol{x}_0)+\boldsymbol{g}(\boldsymbol{x}_0)\cdot\boldsymbol{u}_0$. Depending on the control allocation, the two approaches might result in different solutions for $\boldsymbol{u}$, because for model-based INDI, one might consider allocating an increment $\Delta\boldsymbol{u}$, such that

$$\begin{cases}\boldsymbol{g}(\boldsymbol{x}_0)\cdot\Delta\boldsymbol{u}=\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_{\text{OBM}}\\ \boldsymbol{u}=\boldsymbol{u}_0+\Delta\boldsymbol{u}\end{cases} \tag{57}$$

Case 3. MIMO: $n_u=n_y$

Input non-affine: Here for NDI, the nonlinear control allocation problem

$$\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})=\boldsymbol{v}_{\text{des}}-\boldsymbol{f}(\boldsymbol{x}_0) \tag{58}$$

needs to be solved. This might be not easy if the nonlinear function $\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})$ cannot be solved analytically for $\boldsymbol{u}$. Sometimes, it might be possible to find a nonlinear mapping that defines a new "virtual" input that is input affine.[61,28, p. 249ff.]

The model-based INDI, however, is easily applicable due to the input affinity resulting from the Taylor approximation, which will lead to the inversion law

$$\boldsymbol{u}=\boldsymbol{B}_v^{-1}\cdot(\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_{\text{OBM}})+\boldsymbol{u}_0 \tag{59}$$

with $\dot{\boldsymbol{y}}_{\text{OBM}}=\boldsymbol{f}(\boldsymbol{x}_0)+\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u}_0)$ and $\boldsymbol{B}_v=\partial\boldsymbol{g}(\boldsymbol{x}_0,\boldsymbol{u})/\partial\boldsymbol{u}|_{\boldsymbol{u}=\boldsymbol{u}_0}$.

Note, if an inversion can be found for NDI, that inversion will be exact, while the model-based INDI law is only an approximation.

Case 4. MIMO: $n_u>n_y$

Input non-affine: NDI can become even more challenging than in the previous case and potentially is only solvable if n_u-n_y additional nonlinear constraints are found, that allow a full nonlinear inversion. For model-based INDI, on the other hand, the following approximated linear (!) control allocation problem may be solved by utilizing an appropriate control allocation

$$\boldsymbol{B}_v\cdot\boldsymbol{u}=\boldsymbol{v}_{\text{des}}-\dot{\boldsymbol{y}}_{\text{OBM}}+\boldsymbol{B}_v\cdot\boldsymbol{u}_0 \tag{60}$$

with $\dot{\boldsymbol{y}}_{\mathrm{OBM}} = \boldsymbol{f}(\boldsymbol{x}_0) + \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_0)$ and $\boldsymbol{B}_v = \partial \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u})/\partial \boldsymbol{u}|_{\boldsymbol{u}=\boldsymbol{u}_0}$. Again, this also may be solved incrementally, such that

$$\begin{cases} \boldsymbol{B}_v \cdot \Delta \boldsymbol{u} & = \boldsymbol{v}_{\mathrm{des}} - \dot{\boldsymbol{y}}_{\mathrm{OBM}} \\ \boldsymbol{u} & = \boldsymbol{u}_0 + \Delta \boldsymbol{u} \end{cases} \tag{61}$$

8. INDI – in general case

The previous examples and different derivations of the INDI inversion law have been based on SISO scalar systems with a relative degree of one in order to focus on the underlying concepts. In this section, we want to generalize the derivation of the INDI inversion law to input non-affine, nonlinear, and time-invariant MIMO systems. The derivation loosely follows.[5] Considered are systems of the form

$$\begin{cases} \dot{\boldsymbol{x}} & = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \\ \boldsymbol{y} & = \boldsymbol{h}(\boldsymbol{x}) \end{cases} \tag{62}$$

with states $\boldsymbol{x} \in \mathbb{R}^{n_x}$, outputs $\boldsymbol{y} \in \mathbb{R}^{n_y}$, and (control) inputs $\boldsymbol{u} \in \mathbb{R}^{n_u}$. We assume $n_y \leqslant n_u$, such that the system is fully actuated or overactuated.

Each output y_i has associated a relative degree r_i, which defines how many times it has to be differentiated with respect to time t until the control inputs $\boldsymbol{u}$ can directly impact that respective derivative $y_i^{(r_i)}$. Hence,

$$\frac{\partial y_i^{(r_i)}}{\partial \boldsymbol{u}} \neq \boldsymbol{0}, \quad \frac{\partial y_i^{(k)}}{\partial \boldsymbol{u}} = \boldsymbol{0} \tag{63}$$

for $k \in [0, 1, \ldots, r_i - 1]$. Since these partial derivatives generally depend on state $\boldsymbol{x}$ and input $\boldsymbol{u}$, the relative degree r_i of an output y_i may change during operation.

For example, for transition aircraft, the relative degree of the vertical acceleration will change depending on whether the vehicle is thrust-born or wing-born. When thrust-borne, thrust directly influences the force and, consequently, the acceleration. When wing-borne, the control surfaces cannot directly influence the acceleration. Instead, the surfaces will generate a moment, thus an angular acceleration, leading to a rotational rate, which will change the angle of attack. This change in the angle of attack will finally influence the aerodynamic forces. Hence, the relative degree is larger when the vehicle is wing-borne.

However, we assume that the relative degrees of the outputs remain constant and do not change over time and/or envelope or operation. On system level, the vectorial relative degree is defined as $\boldsymbol{r} = \left[r_1, r_2, \ldots, r_{n_y}\right]^{\mathrm{T}}$ in Ref. 61.p. 220.

We redefine the time derivates $y_i^{(r_i)}$ as pseudo controls v_i, and their dependency on states and inputs as $F_i(\boldsymbol{x}, \boldsymbol{u})$, such that one individual pseudo control is given by

$$v_i = F_i(\boldsymbol{x}, \boldsymbol{u}) = y_i^{(r_i)} \tag{64}$$

For all outputs combined we write in vector notation

$$\boldsymbol{v} = \boldsymbol{F}(\boldsymbol{x}, \boldsymbol{u}) = \boldsymbol{a}_v(\boldsymbol{x}) + \boldsymbol{b}_v(\boldsymbol{x}, \boldsymbol{u}) \tag{65}$$

with state-only dependent part $\boldsymbol{a}_v(\boldsymbol{x})$ and control input dependent part $\boldsymbol{b}_v(\boldsymbol{x}, \boldsymbol{u})$. The following concepts are introduced based on a hypothetical perfect inversion. A perfect inversion of the nonlinearity given in Eq. (65) could be achieved if $\boldsymbol{F}(\boldsymbol{x}, \boldsymbol{u})$ would be known exactly and could explicitly be solved for $\boldsymbol{u}$. In such case, the "perfect inversion law" would be given as

$$\boldsymbol{u} = \boldsymbol{F}^{-1}(\boldsymbol{x}, \boldsymbol{v}) \tag{66}$$

and a desired $\boldsymbol{v}_{\mathrm{des}}$ could be imposed such that

$$\begin{cases} \boldsymbol{v} = \boldsymbol{F}\left(\boldsymbol{x}, \boldsymbol{F}^{-1}(\boldsymbol{x}, \boldsymbol{v}_{\mathrm{des}})\right) \\ \boldsymbol{v} = \boldsymbol{v}_{\mathrm{des}} \end{cases} \tag{67}$$

The combined system, consisting of the system to be controlled Eq. (62) and its inversion Eq. (66), has inputs $\boldsymbol{v}_{\mathrm{des}}$ and outputs $\boldsymbol{y}$. Hence, $\boldsymbol{v}$ are denoted as pseudo controls since they act as new inputs into the combined system. We distinguish $\boldsymbol{v}_{\mathrm{des}}$, which is the desired pseudo control, and $\boldsymbol{v}$, which is the true system reaction. In case of an imperfect inversion, $\boldsymbol{v}_{\mathrm{des}}$ and $\boldsymbol{v}$ may differ to some extent.

We obtain each output y_i by integrating v_i (which is by definition equal to $y_i^{(r_i)}$), r_i times, hence we can represent this as a chain of r_i integrators. For the full system, we will then have $r = r_1 + r_2 + \ldots + r_{n_y}$ number of integrators. The system is said to have a "full relative degree" if $r = n_x$. Then there is the same number of **new** states represented by the integrators, as the number of original system states $\boldsymbol{x}$. The set of new states (also called "external states" as in Ref. 31 and often denoted as $\boldsymbol{\xi}$ as in Refs. 28,61) is a valid and invertible state transformation.[28]

If the system does not have a full relative degree, $r < n_x$, a perfect inversion would still result in a combined system consisting purely of chains of integrators. However, now, the number of integrators in these chains (still called external states $\boldsymbol{\xi}$) is less than the number of states $\boldsymbol{x}$ of the original system. Of course, the "missing" states do not disappear, but they, in fact, become unobservable in the combined system. These states are called "internal states" as in Ref. 31 and are often denoted as $\boldsymbol{\eta}$ as in Refs. 28,61.

Assessing the stability of the internal states' governing "internal dynamics" is an important consideration for such systems. Possibilities in doing so are dynamic extension,[61] stability assessment of the zero dynamics,[61] stabilization through outer-loops etc.

At this point, we will turn the focus to INDI, considering the transformed system representation given by equation Eq. (65).

In the current operating point $(\boldsymbol{u}_0, \boldsymbol{x}_0)$, which means the physical realization of the variables at a distinctive point in time, the plant pseudo control results in $\boldsymbol{v}_0$. Hence the following equation is satisfied

$$\boldsymbol{v}_0 = \boldsymbol{F}(\boldsymbol{x}_0, \boldsymbol{u}_0) = \boldsymbol{a}_v(\boldsymbol{x}_0) + \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_0) \tag{68}$$

If instead we want to achieve a desired pseudo control $\boldsymbol{v}_{\mathrm{des}}$, the control inputs represent the degrees of freedom to be changed. Let's denote the ideal control input as $\boldsymbol{u}_{\mathrm{ideal}}$, which would satisfy

$$\boldsymbol{v}_{\mathrm{des}} = \boldsymbol{F}(\boldsymbol{x}_0, \boldsymbol{u}_{\mathrm{ideal}}) = \boldsymbol{a}_v(\boldsymbol{x}_0) + \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_{\mathrm{ideal}}) \tag{69}$$

It is like asking the question "what control input $\boldsymbol{u}_{\mathrm{ideal}}$ would be required, to have a desired pseudo control $\boldsymbol{v}_{\mathrm{des}}$ in this very moment?". Subtracting Eq. (68) from Eq. (69) leads to

$$\begin{aligned} \boldsymbol{v}_{\mathrm{des}} - \boldsymbol{v}_0 & = \boldsymbol{F}(\boldsymbol{x}_0, \boldsymbol{u}_{\mathrm{ideal}}) - \boldsymbol{F}(\boldsymbol{x}_0, \boldsymbol{u}_0) \\ & = \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_{\mathrm{ideal}}) - \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_0) \end{aligned} \tag{70}$$

where the state only dependent term $\boldsymbol{a}_v(\boldsymbol{x}_0)$ cancels completely. Now we have to distinguish two cases:

Case 1. Input affine systems.
Case 2. Input non-affine systems.

If the system is input affine (Case 1), Eq. (70) results as linear equation

$$\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0 = \boldsymbol{B}_v(\boldsymbol{x}_0) \cdot \boldsymbol{u}_{\text{ideal}} - \boldsymbol{B}_v(\boldsymbol{x}_0) \cdot \boldsymbol{u}_0 \tag{71}$$

for which the ideal $\boldsymbol{u}_{\text{ideal}}$ in fact can be found without approximation. If the system is square, the resulting inversion law yields

$$\boldsymbol{u}_{\text{cmd}} = \boldsymbol{B}_v^{-1}(\boldsymbol{x}_0) \cdot (\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0) + \boldsymbol{u}_0 \tag{72}$$

for which $\boldsymbol{u}_{\text{cmd}}$ corresponds to the ideal control input $\boldsymbol{u}_{\text{ideal}}$. If the system is overactuated, control allocation algorithms have to be utilized. For example, using the pseudoinverse then results in the inversion law

$$\boldsymbol{u}_{\text{cmd}} = \boldsymbol{B}_v^{\dagger}(\boldsymbol{x}_0) \cdot (\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0) + \boldsymbol{u}_0 \tag{73}$$

If the system is input non-affine (Case 2), one could try to solve the nonlinear equation

$$\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0 = \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_{\text{ideal}}) - \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_0) \tag{74}$$

for $\boldsymbol{u}_{\text{ideal}}$ through iterative methods or by other means, but in practice that might be difficult, especially for overactuated systems. Therefore, the nonlinear function $\boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u})$ is approximated and replaced by a linear relation around $\boldsymbol{u}_0$, given by

$$\tilde{\boldsymbol{b}}_v(\boldsymbol{x}_0, \boldsymbol{u}) = \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_0) + \left.\frac{\partial \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_0} \cdot (\boldsymbol{u} - \boldsymbol{u}_0) \tag{75}$$

similar to the derivation of sensory NDI of Refs. 58,60. Due to the approximation, one does not find an ideal $\boldsymbol{u}_{\text{ideal}}$ anymore, and hence it is relabelled as a "required" $\boldsymbol{u}_{\text{req}}$, which according to the approximated equation would lead to the desired pseudo control. Inserting the approximation Eq. (75) in Eq. (74) leads to

$$\begin{aligned} \boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0 &= \tilde{\boldsymbol{b}}_v(\boldsymbol{x}_0, \boldsymbol{u}_{\text{req}}) - \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u}_0), \\ &= \left.\frac{\partial \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_0} \cdot (\boldsymbol{u}_{\text{req}} - \boldsymbol{u}_0) \end{aligned} \tag{76}$$

Substituting the partial derivative as

$$\boldsymbol{B}_v(\boldsymbol{x}_0, \boldsymbol{u}_0) = \left.\frac{\partial \boldsymbol{b}_v(\boldsymbol{x}_0, \boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\boldsymbol{u}=\boldsymbol{u}_0} \tag{77}$$

results in an equation that is similar to the input affine Eq. (71).

$$\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0 = \boldsymbol{B}_v(\boldsymbol{x}_0, \boldsymbol{u}_0) \cdot (\boldsymbol{u}_{\text{req}} - \boldsymbol{u}_0) \tag{78}$$

and can be solved by appropriate control allocation methods again. The resulting inversion law for a square system results in

$$\boldsymbol{u}_{\text{cmd}} = \boldsymbol{B}_v^{-1}(\boldsymbol{x}_0, \boldsymbol{u}_0) \cdot (\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0) + \boldsymbol{u}_0 \tag{79}$$

whereas for an overactuated system the inversion law may follow as

$$\boldsymbol{u}_{\text{cmd}} = \boldsymbol{B}_v^{\dagger}(\boldsymbol{x}_0, \boldsymbol{u}_0) \cdot (\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0) + \boldsymbol{u}_0 \tag{80}$$

if the pseudoinverse is chosen as control allocation for instance.

In both inversion laws, the required control input $\boldsymbol{u}_{\text{req}}$ has already been renamed to $\boldsymbol{u}_{\text{cmd}}$. The matrix $\boldsymbol{B}_v(\boldsymbol{x}_0, \boldsymbol{u}_0)$ is the current sensitivity of the pseudo control with respect to the control inputs, which may depend on current states and/or control inputs. This matrix is also denoted as "Control Effectiveness". The terms $\boldsymbol{u}_0$ and $\boldsymbol{v}_0$ are measurements and/or estimates of the current control input and pseudo control values. Depending on the dimensions and actuator saturations, the inversion of $\boldsymbol{B}_v$ is either a standard matrix inverse $\boldsymbol{B}_v^{-1}$, a pseudoinverse $\boldsymbol{B}_v^{\dagger}$ or a more sophisticated control allocation method. The next session will discuss in more detail the implications of actuator dynamics.

Note 7. Note that for a MIMO system with a relative degree higher than one, the INDI control law can also be derived using Lie-derivatives as, for example, in Refs. 62,63.

Note 8. For overactuated systems, the resulting closed-loop pseudo control dynamics may depend on the control allocation, which we want to explain in the context of an overactuated input affine MIMO system given by

$$\boldsymbol{v} = \boldsymbol{B}_v(\boldsymbol{x}) \cdot \boldsymbol{u} \tag{81}$$

The solution $\boldsymbol{u}_{\text{cmd}}$ to the overactuated control allocation problem may in general be written as

$$\boldsymbol{u}_{\text{cmd}} = \boldsymbol{B}_v^I(\boldsymbol{x}_0, t_0) \cdot (\boldsymbol{v}_{\text{des}} - \boldsymbol{v}_0) + \boldsymbol{u}_0 \tag{82}$$

where $\boldsymbol{B}_v^I(\boldsymbol{x}_0, t_0)$ is a generalized inverse matrix depending on the control allocation, such that at any time holds $\boldsymbol{B}_v(\boldsymbol{x}_0) \cdot \boldsymbol{B}_v^I(\boldsymbol{x}_0, t_0) = \boldsymbol{I}$. In contrast to the fast $\boldsymbol{v}$ dynamics, we assume the control effectiveness $\boldsymbol{B}_v(\boldsymbol{x})$ to depend only on slowly changing (state) variables, such as dynamic or static pressure. Furthermore, we assume the dependence on the states to be such that $\boldsymbol{B}_v(\boldsymbol{x})$ will change only slowly over time. Hence, in the following derivation we can assume $\boldsymbol{B}_v$ to be a quasi-constant, time invariant matrix.

In general, due to the overactuation, $\boldsymbol{B}_v$ as a broad matrix has a nullspace and there is no unique inverse for it - in fact there are infinitely many (generalized) inverses possible in Ref. 64. These additional degrees of freedom can be utilized by the control allocation in order to fulfill secondary objectives. Therefore, the exact realization of $\boldsymbol{B}_v^I(\boldsymbol{x}_0, t_0)$ depends on the control allocation and might change from one time step to another, hence, it is depending on time. However, to simplify the following derivations, we assume the $\boldsymbol{B}_v^I(\boldsymbol{x}_0, t_0)$ matrix to be a constant matrix $\boldsymbol{B}_{v,1}^I(\boldsymbol{x}_0)$, which might change at one distinctive point in time to another matrix $\boldsymbol{B}_{v,2}^I(\boldsymbol{x}_0)$. Due to the same reasons as mentioned above, the state dependency of $\boldsymbol{B}_v^I(\boldsymbol{x}_0)$ is assumed to be quasi-constant as well, such that we will write $\boldsymbol{B}_v^I$ in the following derivations.

Assuming linear control effector dynamics represented by the diagonal transfer matrix $\boldsymbol{G}(s)$ and considering continuous time (such that $\boldsymbol{u}_0 = \boldsymbol{u}$ and $\boldsymbol{v}_0 = \boldsymbol{v}$), then the resulting control inputs in Laplace domain can be calculated as

$$\begin{cases} \boldsymbol{U} = \boldsymbol{G} \cdot \boldsymbol{U}_{\text{cmd}} \\ \boldsymbol{U} = \boldsymbol{G} \cdot \left(\boldsymbol{B}_v^I \cdot (\mathscr{V}_{\text{des}} - \mathscr{V}) + \boldsymbol{U}\right) \\ (\boldsymbol{I}_u - \boldsymbol{G}) \cdot \boldsymbol{U} = \boldsymbol{G} \cdot \boldsymbol{B}_v^I \cdot (\mathscr{V}_{\text{des}} - \mathscr{V}) \\ \boldsymbol{U} = (\boldsymbol{I}_u - \boldsymbol{G})^{-1} \cdot \boldsymbol{G} \cdot \boldsymbol{B}_v^I \cdot (\mathscr{V}_{\text{des}} - \mathscr{V}) \end{cases} \tag{83}$$

where the Laplace operator (s) has been omitted for better readability and where $\boldsymbol{I}_u$ denotes the identity matrix with num-

ber of rows/columns corresponding to the number of elements in $\boldsymbol{u}$. The expression $(\boldsymbol{I}_u - \boldsymbol{G}(s))^{-1} \cdot \boldsymbol{G}(s)$ in fact represents a diagonal transfer matrix as well. Substituted by

$$\widetilde{\boldsymbol{G}}(s) = (\boldsymbol{I}_u - \boldsymbol{G}(s))^{-1} \cdot \boldsymbol{G}(s) \tag{84}$$

this will simplify the expression for the control input to

$$\boldsymbol{U} = \widetilde{\boldsymbol{G}} \cdot \boldsymbol{B}_v^I \cdot (\boldsymbol{\mathcal{V}}_{\text{des}} - \boldsymbol{\mathcal{V}}) \tag{85}$$

Inserted into the input affine transformed dynamics from Eq. (81) (in Laplace domain) yields

$$\boldsymbol{\mathcal{V}}(s) = \boldsymbol{B}_v \cdot \widetilde{\boldsymbol{G}}(s) \cdot \boldsymbol{B}_v^I \cdot (\boldsymbol{\mathcal{V}}_{\text{des}}(s) - \boldsymbol{\mathcal{V}}(s)) \tag{86}$$

In the following, we substitute the resulting transfer matrix as

$$\boldsymbol{G}_B(s) = \boldsymbol{B}_v \cdot \widetilde{\boldsymbol{G}}(s) \cdot \boldsymbol{B}_v^I \tag{87}$$

Using the just introduced substitution, the closed loop dynamics can be written as

$$\begin{cases} \boldsymbol{\mathcal{V}} = \boldsymbol{G}_B \cdot (\boldsymbol{\mathcal{V}}_{\text{des}} - \boldsymbol{\mathcal{V}}) \\ (\boldsymbol{I}_v + \boldsymbol{G}_B) \cdot \boldsymbol{\mathcal{V}} = \boldsymbol{G}_B \cdot \boldsymbol{\mathcal{V}}_{\text{des}} \end{cases} \tag{88}$$

where $\boldsymbol{I}_v$ denotes the identity matrix with number of rows/-columns corresponding to the number of elements in $\boldsymbol{v}$. That equation can be solved to obtain the resulting closed-loop pseudo control response

$$\boldsymbol{\mathcal{V}} = (\boldsymbol{I}_v + \boldsymbol{G}_B)^{-1} \cdot \boldsymbol{G}_B \cdot \boldsymbol{\mathcal{V}}_{\text{des}} \tag{89}$$

Now we can distinguish two cases:

Case 1. The control effectors have all the same dynamics.
Case 2. The control effectors have different dynamics.

Case 1. If all control effectors have equal dynamics, all diagonal entries in $\boldsymbol{G}(s)$ are identical and will be denoted as $g(s)$. Furthermore, the diagonal entries in $\widetilde{\boldsymbol{G}}(s)$ are identical as well and will be denoted as $\tilde{g}(s)$. We can then write the resulting transfer matrix $\boldsymbol{G}_B(s)$ as

$$\begin{aligned} \boldsymbol{G}_B(s) &= \boldsymbol{B}_v \cdot \tilde{g}(s) \cdot \boldsymbol{I}_u \cdot \boldsymbol{B}_v^I \\ &= \tilde{g}(s) \cdot \boldsymbol{I}_v \cdot \boldsymbol{B}_v \cdot \boldsymbol{B}_v^I \\ &= \tilde{g}(s) \cdot \boldsymbol{I}_v \end{aligned} \tag{90}$$

Key takeaway is, that here the matrix product $\boldsymbol{B}_v \cdot \boldsymbol{B}_v^I$ cancels, and the resulting closed loop transfer function therefore does not depend on $\boldsymbol{B}_v^I$ any longer

$$\begin{aligned} \boldsymbol{\mathcal{V}} &= \left(1 + \tilde{\tilde{g}}(s)\right)^{-1} \cdot \tilde{\tilde{g}}(s) \cdot \boldsymbol{I}_v \cdot \boldsymbol{\mathcal{V}}_{\text{des}} \\ &= g(s) \cdot \boldsymbol{I}_v \cdot \boldsymbol{\mathcal{V}}_{\text{des}} \end{aligned} \tag{91}$$

This means that the resulting transfer function for each v channel is precisely the control effector dynamics, all channels are decoupled, and there is no dependency on what kind of inverse $\boldsymbol{B}_v^I$ is used in the control allocation - even if this inverse changes.

Case 2. If the control effector dynamics are not all equal, $\boldsymbol{G}_B(s)$is not diagonal and in general no good-natured cancellation will happen. Basically, in eq. (Eq. (87)), the matrices $\boldsymbol{B}_v$ and $\boldsymbol{B}_v^I$ do not cancel, because of the diagonal transfer matrix $\widetilde{\boldsymbol{G}}(s)$ (with non-equal diagonal entries) in between them. That has two implications: For square MIMO system (which are not overactuated, and for which the inverse $\boldsymbol{B}_v^I$ is unique), the resulting transfer function as described in eq. (Eq. (89)) results in a fully set transfer matrix $(\boldsymbol{I}_v + \boldsymbol{G}_B)^{-1} \cdot \boldsymbol{G}_B$ depending on the exact realization of $\boldsymbol{B}_v$ and depending on the control effector dynamics $\boldsymbol{G}(s)$. "Fully set" means, that in general cross couplings in between the different pseudo control channels are introduced, such that for example a desired $v_{\text{des},1}$ will also have an impact on a resulting v_2. Furthermore, for overactuated systems, the $\boldsymbol{B}_v^I$ matrix is not unique, such that the resulting closed-loop dynamics depends on $\boldsymbol{B}_v^I$ as well. Since that matrix is mainly dominated by the control allocation and might change over time, this concludes our statement that the resulting closed loop pseudo control dynamics for overactuated systems for classical INDI might change over time and is depending on the control allocation, if the control effector dynamics are not all equal.

9. Extended INDI (E-INDI) and Actuator NDI (ANDI) – Consideration of actuator dynamics

In the INDI derivations so far, actuator dynamics were neglected, and the commanded u was assumed to be reached instantaneously. For real applications, however, actuator dynamics will cause the commanded Δu to build up over time rather than being achieved immediately. Hence, the desired pseudo control increment Δv will also be delayed. This fact will deteriorate the phase and gain of the pseudo control response, as, for example, visualized in general in Ref. 65, and in the context of INDI in Ref. 66.

In 2005, Ref. 65 has described actuator compensation for first-order, second-order, and second-order with a single zero linear systems. The derivations are performed in discrete time, and the actuator compensation results in modifying the commanded u after control allocation. Ref. 67 has later discussed discrete linear actuator dynamics of arbitrary order. They consider these actuator dynamics within a dynamic control allocation, not after, by including the actuator information within the LMI problem solved in the allocator. The algorithms presented in Ref. 65 and Ref. 67 are independent of dynamic inversion and focus on constraint control allocation in the presence of actuator dynamics. Their feasibility is demonstrated through control allocator simulations, and in Ref. 67, also through closed-loop simulations with a linear plant and linear controller setup.

In 2018, Ref. 66 brought compensation of actuator dynamics in the context of INDI. Using the continuous frequency domain representation of first-order and second-order actuators, they derive gains to multiply by the output of the control allocation, compensating for the actuator dynamics. The procedure resembles the derivations of Ref. 65, but they are set in the context of INDI, and the resulting gains are simpler since they consider continuous time. They apply their compensation after control allocation, before sending the command to the actuator.

9.1. E-INDI

In 2019, Ref. 68 introduced a continuous reformulation of the INDI law, in which they can incorporate actuator dynamics directly within the control allocation. The procedure to obtain the necessary transfer functions generally resembles Ref. 65 and Ref. 66, providing additional insights via graphical repre-

sentations. The reformulation furthermore allows the decoupled imposing of desired pseudo control dynamics. As Ref. 3 shows, the pseudo control loop, meaning the closed-loop dynamics with an INDI controller from v_{des} to v_0, results in the actuator dynamics. However, if multiple actuators are involved, potentially with different bandwidths and orders, the response will be an allocation-dependent combination of the actuator dynamics involved as Note 8 elaborated.

The results of Ref. 68 allow the decoupling of the pseudo control and actuator loops, such that desired pseudo control dynamics can be specified, which gives a straightforward way to design outer-loops. The upcoming paragraphs briefly explain the outcomes of Ref. 68. First of all, note that linear second-order actuator dynamics of the form $\omega_0^2/\left(s^2+2\zeta\omega_0\cdot s+\omega_0^2\right)$ can be formulated as an inner-loop and an outer-loop as visualized in Fig. 5 with $\omega_{\text{IL}}=2\zeta\omega_0$ and $\omega_{\text{OL}}=\omega_0/(2\zeta)$.[68,69]

In the case of linear first-order actuators, the inner-loop transfer function simply becomes 1, and ω_{OL} will correspond to $1/T$, where T is the time constant. When drawing the resulting actuator block diagram together with the calculation of the INDI command $u_{\text{cmd}}=\Delta u_{\text{cmd}}+u_0$, we see that the addition of u_0 to Δu_{cmd} will compensate the u feedback within the actuator dynamics, as depicted in Fig. 6. The transfer behavior between Δu_{cmd} and $\dot{u}$ for a first-order actuator is given by in the algebraic relation

$$\dot{u}=\omega_{\text{OL}}\cdot\Delta u_{\text{cmd}} \tag{92}$$

That means, we can command a desired actuator rate $\dot{u}_{\text{cmd}}$ and by multiplying it with $1/\omega_{\text{OL}}$ we obtain the Δu_{cmd}. Hence, as depicted in Fig. 7, the rate command follows as

$$\Delta u_{\text{cmd}}=1/\omega_{\text{OL}}\cdot\dot{u}_{\text{cmd}} \tag{93}$$

The extended INDI law[68] takes these results into account. We derive it in the following, on an input affine, scalar SISO system, to illustrate the concept. The mentioned system with invertible $g(x)$ is given as

$$\begin{cases}\dot{x} &= f(x)+g(x)\cdot u\\ y &= x\end{cases} \tag{94}$$

where x is the state, u is the control input vector and y is the output. If the system has a relative degree of one with respect to u, then for nominal INDI, the pseudo control is defined as $v=\dot{y}=\dot{x}$ and given by

$$v=f(x)+g(x)\cdot u=F(x,u) \tag{95}$$

For the extended INDI derivation, we take another time derivative of Eq. (95), which results in

$$\begin{cases}\dot{v} &= \frac{\partial F(x,u)}{\partial x}\cdot\dot{x}+\frac{\partial F(x,u)}{\partial u}\cdot\dot{u}\\ \dot{v} &= A_v\cdot\dot{x}+B_v\cdot\dot{u}\end{cases} \tag{96}$$

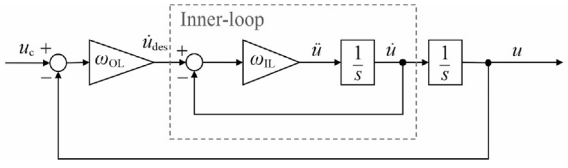


Fig. 5 Linear second-order actuators as inner-loop and outer-loop representation.[68,69]

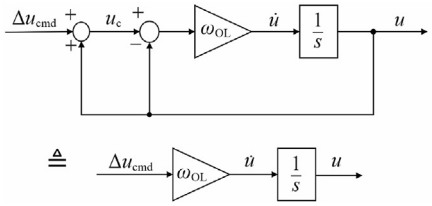


Fig. 6 Compensation of u feedback in actuator dynamics.

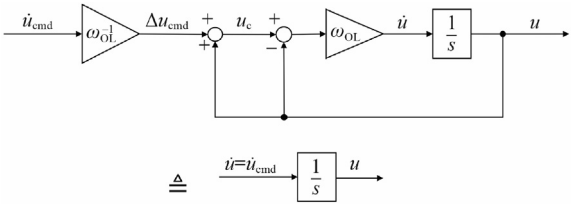


Fig. 7 Inversion of actuator dynamics.

Depending on the system, the term $A_v\cdot\dot{x}$ may be omitted, and the resulting approximated system dynamics used for the inversion law follows as

$$\dot{v}\approx B_v\cdot\dot{u} \tag{97}$$

This result may be used twofold. First, one could define the desired dynamics for the pseudo control, following the same principle as the previously described actuator transfer functions (compare Eq. (92)). For first-order desired dynamics, this would lead to

$$\dot{v}=\omega_{v,\text{des}}\cdot\Delta v \tag{98}$$

where $\omega_{v,\text{des}}$ is the desired pseudo control bandwidth. Then both equations Eq. (92) and Eq. (98) may be inserted in Eq. (97), to form the modified inversion law

$$\Delta v=\left(1/\omega_{v,\text{des}}\right)\cdot B_v\cdot\omega_{\text{OL}}\cdot\Delta u_{\text{cmd}} \tag{99}$$

which in turn is solved for Δu_{cmd} using an appropriate control allocation. This especially has advantages for MIMO systems with multiple actuators with different bandwidths because it allows the specification of desired dynamics for each pseudo control channel separately.

To illustrate the implementable extended INDI inversion law, in the following the SISO inversion law is shown. It follows as

$$\Delta u_{\text{cmd}}=\left(B_v\cdot\omega_{\text{OL}}\right)^{-1}\cdot\omega_{v,\text{des}}\cdot\Delta v \tag{100}$$

where ω_{OL} again describes the effective outer loop bandwidth of the control effector and where $\omega_{v,\text{des}}$ describes the design parameter to specify the desired pseudo control bandwidth. In MIMO systems, both ω_{OL} and $\omega_{v,\text{des}}$ are diagonal matrices with appropriate sizes, where the diagonal elements represent the bandwidths of the respective channels. For overactuated systems, appropriate control allocation algorithms have to be utilized to solve the inversion.

A second way to utilize Eq. (97) is to stay on the $\dot{v}$ level and command a desired $\dot{v}$ to the control allocation, which allocates a necessary $\dot{u}$, which in turn is translated into a required Δu_{cmd} following Eq. (93). This desired $\dot{v}$ would be computed in an

outer v control loop, with feedforward and feedback path. The extended INDI SISO inversion law in that case follows as

$$\Delta u_{\text{cmd}} = (B_v \cdot \omega_{\text{OL}})^{-1} \cdot \dot{v}_{\text{des}} \tag{101}$$

In 2021, Ref. 70, for instance, extended these results of Ref. 68 by shaping the feedforward path of the desired pseudo control dynamics in an integrated reference model that commands $\dot{v}_{\text{ref}}$ to the control allocation and accounts for the state-dependent part over $A_v \cdot \dot{x}_{\text{ref}}$. We address this topic in more detail in the second part of this survey[30] in the section related to reference models. For the sake of completeness, the extended INDI SISO inversion law then results in

$$\Delta u_{\text{cmd}} = (B_v \cdot \omega_{\text{OL}})^{-1} \cdot (\dot{v}_{\text{des}} - A_v \cdot \dot{x}_{\text{ref}}) \tag{102}$$

Note 9. In Note 8 we showed that in case of multiple effectors with different dynamics, INDI does not decouple the pseudo-controls anymore. This means that the pseudo-controls cannot be specified independently by respective $\boldsymbol{v}_{\text{des}}$ anymore, and cross-couplings might arise.

This is not the case for E-INDI. To show that E-INDI accounts for the effector dynamics such that the v dynamics are decoupled and can be specified independently, we consider, similar to Note 8, the overactuated input affine MIMO system given by Eq. (81).

Assuming again $\boldsymbol{B}_v$ to be constant as it is only slowly changing with the slow varying states $\boldsymbol{x}$, then the pseudo-control derivative results as

$$\dot{\boldsymbol{v}} = \boldsymbol{B}_v \cdot \dot{\boldsymbol{u}} = \boldsymbol{B}_v \cdot \boldsymbol{\omega}_{\text{OL}} \cdot (\boldsymbol{u}_{\text{cmd}} - \boldsymbol{u}_0) \tag{103}$$

where first-order effector dynamics are assumed and $\boldsymbol{\omega}_{\text{OL}}$ is a diagonal matrix with diagonal entries corresponding to the bandwidths of the respective effectors. The E-INDI inversion law in that case follows as (compare to Eq. (81)

$$\boldsymbol{u}_{\text{cmd}} = (\boldsymbol{B}_v \cdot \boldsymbol{\omega}_{\text{OL}})^I \cdot \dot{\boldsymbol{v}}_{\text{des}} + \boldsymbol{u}_0 \tag{104}$$

where $(\boldsymbol{B}_v \cdot \boldsymbol{\omega}_{\text{OL}})^I$ is a generalized inverse matrix depending on the control allocation, such that at any time holds $(\boldsymbol{B}_v \cdot \boldsymbol{\omega}_{\text{OL}})(\boldsymbol{B}_v \cdot \boldsymbol{\omega}_{\text{OL}})^I = \boldsymbol{I}_u$, where $\boldsymbol{I}_u$ denotes the identity matrix with number of rows/columns corresponding to the number of elements in $\boldsymbol{u}$. If we insert the E-INDI law Eq. (104) into Eq. (103), we see that $\dot{\boldsymbol{v}} = \dot{\boldsymbol{v}}_{\text{des}}$. The resulting $\boldsymbol{v}$ dynamics are decoupled, and we can specify independently the respective desired $\dot{\boldsymbol{v}}_{\text{des}}$.

Note that in Eq. (103), we assumed linear first-order actuators. In fact, the cancellation is only exact in that case and if the actuator bandwidth ω_{OL} is known. For linear second-order dynamics, E-INDI can still be applied using the "effective" outer loop bandwidth of the control effector, see Fig. 5. However, the decoupling will not be exact anymore, but in general better than without the E-INDI actuator compensation.

Note 10. In case the actuator/control effector dynamics are nonlinear, one may also utilize this result to allocate a desired $\dot{v}$ into a necessary $\dot{u}$ and then calculate the required effector command through a nonlinear relation. For instance, an electric motor driving a rotor may follow a quadratic relation

$$\dot{\omega}_p = K_1 \cdot \omega_p^2 + K_2 \cdot \omega_p + K_3 \cdot u_{\text{cmd}} \tag{105}$$

in which case ω_p is used as virtual controls, as it is the physical "effect" that is used as control input for the system control. Hence, the desired $\dot{v}$ is allocated into a necessary $\dot{\omega}_p$. However, the actual command to be sent to the motor is u_{cmd}. Given a good understanding of the effector parameters K_1, K_2 and K_3 (through identification), and given an estimate or measurement of the current propeller rotational rate ω_p, one could solve for the required u_{cmd}, which causes the necessary $\dot{\omega}_p$ that would cause a desired $\dot{v}$ on the system (for instance a force or moment reaction).

Note 11. Refs. 1,58 give another example of a virtual control choice by

$$u = \omega_p^2 \tag{106}$$

In Ref.1 a desired $\dot{v}$ is allocated into a necessary $\dot{u}$ and then solved for $\dot{\omega}_p$ using the relation $\dot{u} = 2\omega_p \cdot \dot{\omega}_p$. Finally, $\dot{\omega}_p = \frac{\dot{u}}{2\omega_p}$ is used to calculate the actual $\Delta\omega_{p,\text{cmd}}$ using the actuator dynamics as described by Eq. (93).

9.2. E-INDI: Integrating INDI and actuator control across hardware units

A key finding from E-INDI[68] is that the "increment" control input command Δu in principle represents a direct actuator rate command $\dot{u}$, because the u feedback into the INDI law cancels the intrinsic u feedback of the actuator control loop, as Fig. 8 shows. For real systems, commonly, both of these loops are physically there:

On the one hand, there is the actuator with an actuator controller, to which, for example, a desired position can be commanded. Hence, the actuator intrinsic u feedback is within the module "actuator + actuator controller.".

The second u feedback is utilized within the incremental law $u_{\text{cmd}} = u + \Delta u$. It, therefore, "lives" inside the flight controller (this feedback may be based on a measurement of the actuator position, or based on an onboard model of the actuator, or both). However, the cancellation can never be perfect, as the actuator controller and INDI usually run on different hardware with different sampling rates, and because measurement inaccuracies and delays might be present.

The question arises, whether the aircraft control and actuator control can be considered in one unit instead of staying with the classical separation of flight and actuation control system. Then, in the context of INDI, both actuator feedbacks can be cancelled out perfectly (by avoiding to implement them in the first place).

Indeed, this can and has been done. Ref. 71 followed a similar strategy and directly map the pseudo control increment Δv (in their case, an angular acceleration increment) into a desired actuator current increment, which they command to the actuator controller which accepts current commands.

In case the actuators are commercial-of-the-shelf components, which may not allow direct rate control, Rieger[72] derived several methods of actuator feedforward control to emulate an actuator rate command. However, as mentioned above, both u feedbacks still exist in that case, and the cancellation may never be perfect.

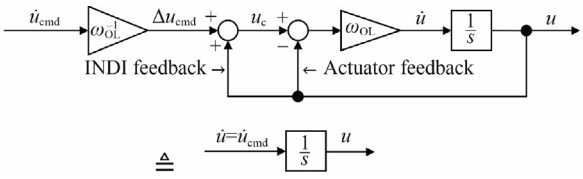


Fig. 8 Compensation of u feedback in actuator dynamics adapted from Ref. 68.

Ref. 73 developed an approach similar to E-INDI, though coming from the actuator control perspective rather than from the INDI perspective. In classical control approaches, the actuator and plant dynamics are considered separately, and separate controllers are developed, which are afterward cascaded. The actuator controller ensures that the commanded actuator positions δ_c are tracked based on actuator position feedback, see the upper block diagram in Fig. 9.

A separate outer-loop controller controls the aircraft dynamics. By considering the flight control and actuator control jointly, Ref. 73 showed that instead of feeding back the actuator position, it is equivalent to feedback the acceleration when mapping the controller gain values appropriately. Fig. 9 depicts the structure in the lower block diagram.

In contrast to actuator position feedback, the acceleration feedback has the advantage of increasing robustness towards uncertainties and disturbances. It is worth to note how this finding aligns with E-INDI, where in the INDI law accelerations are fed back and where in case of perfect cancellation of "INDI u" and "actuator u" feedback (compare Fig. 8), the control law output can be interpreted as $\dot{u}_{\text{cmd}}$. This is very similar to the actuator control part in Fig. 9 in the lower block diagram.

Instead of closing the INDI u feedback loop on the Flight Control Computer (FCC) one could therefore consider to move that loop to the Actuator Control Units (ACU). Then, the FCC commands $\dot{u}_{\text{cmd}}$ to the ACU, where the ACU then contains both "u feedbacks" inside: the positive feedback from the INDI branch, and the negative actuator feedback closing the actuator position loop (compare Fig. 8). In case of a similar actuator control structure as in Fig. 9, one could then consider to omit the u feedback altogether, as they are supposed to cancel anyway. The same principle can also be considered for control effectors in general, such as for example electrical motors, commonly used in multicopter, transition vehicles, etc.

9.3. ANDI

Later in 2022, Ref. 57 elaborated the idea of an extended INDI law further, based on NDI that takes the actuator dynamics into account. In classical NDI, the actuator dynamics are assumed to be much faster than the system dynamics and are therefore neglected. Here, first-order actuator dynamics are appended to the system dynamics, and the NDI inversion law is then formulated for the actuator input. This results in an exact inversion of the complete system, including the actuator dynamics. Their analysis and derivations provide several insights. Using a specific choice of the desired error dynamics, they show that for high actuator bandwidth, the classical INDI law approaches the exact NDI law (that includes actuator dynamics). The derivation is based on a system with relative degree $r \in \mathbb{N}$ given by

$$\begin{cases} \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \\ \boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}) \end{cases} \tag{107}$$

where $\boldsymbol{x} \in \mathbb{R}^{n_x}$ is the system state, $\boldsymbol{u} \in \mathbb{R}^{n_u}$ is the actuator state, $\boldsymbol{y} \in \mathbb{R}^{n_y}$ is the output. Further, the actuator dynamics are given by

$$\dot{\boldsymbol{u}} = \boldsymbol{\Omega}(\boldsymbol{u}_{\text{cmd}} - \boldsymbol{u}) \tag{108}$$

where $\Omega \in \mathbb{R}^{n_u \times n_u}$ is diagonal with the diagonal elements representing the bandwidths ω_a of the respective actuators. For simplicity, we will consider in the following that the system has a relative degree of $r = 1$ and that $n_x = n_u = n_y = 1$. Then

$$\dot{y} = \frac{\partial h(x)}{\partial x} \cdot \dot{x} = \frac{\partial h(x)}{\partial x} \cdot f(x, u) = F(x, u) \tag{109}$$

and

$$\begin{aligned} \ddot{y} &= \tfrac{\partial F(x,u)}{\partial x} \cdot \dot{x} + \tfrac{\partial F(x,u)}{\partial u} \cdot \dot{u} \\ &= F_x(x, u) \cdot \dot{x} + F_u(x, u) \cdot \dot{u} \end{aligned} \tag{110}$$

By inserting the actuator dynamics $\dot{u} = \omega_a \cdot (u_{\text{cmd}} - u)$ into Eq. (110), as well as the pseudo control $v_{\ddot{y}} = \ddot{y}$, Eq. (110) can be solved for u_{cmd}, resulting in the NDI law

$$u_{\text{cmd}} = \frac{1}{F_u(x, u) \cdot \omega_a} \cdot \left(v_{\ddot{y},\text{des}} - F_x(x, u) \cdot \dot{x} \right) + u \tag{111}$$

This resulting command is an exact inversion of the complete system, including the actuator dynamics.

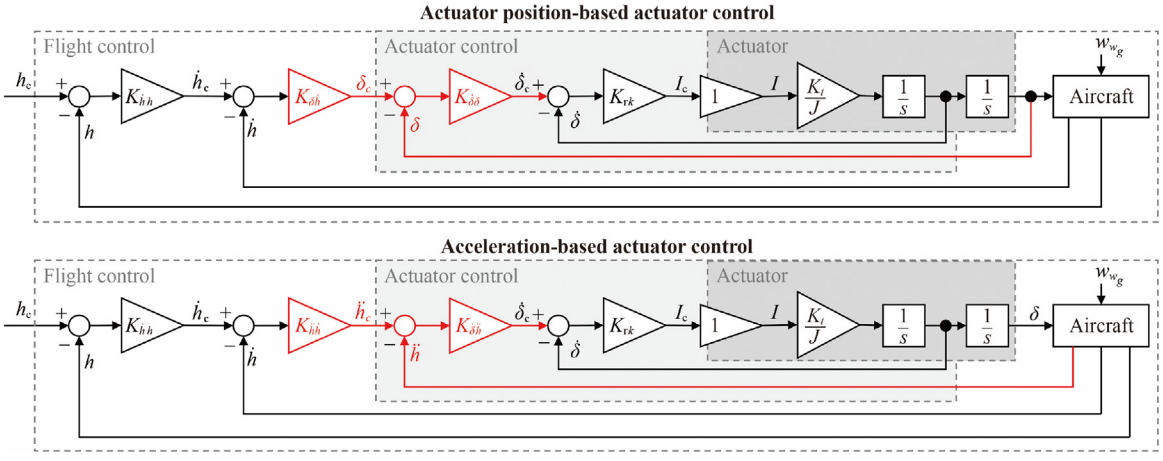


Fig. 9 Flight controller with a separated actuator controller and an acceleration-based actuator control, based on Ref. 73.

An interesting observation can be made by choosing the pseudo control based on a special formulation of the error dynamics. As here the pseudo control $v_{\ddot{y}}$ corresponds to $\ddot{y}$, the dynamics of the error $e = y_{\text{des}} - y$, where y_{des} is the desired output, are specified to be second-order

$$\ddot{e} + k_1 \cdot \dot{e} + k_0 \cdot e = 0 \tag{112}$$

A physically feasible choice for the coefficients is thereby obtained by considering the error dynamics in the Laplace domain and interpreting them as cascaded dynamics with a faster desired inner-loop dynamics or actuator dynamics, specified by ω_i, and the slower desired system error dynamics, ω_e.

$$\begin{aligned} E(s) \cdot (s^2 + k_1 \cdot s + k_0) &= E(s) \cdot (s + \omega_e) \cdot (s + \omega_i) \\ &= E(s) \cdot (s^2 + \omega_e \cdot s + \omega_i \cdot s + \omega_e \cdot \omega_i) \end{aligned} \tag{113}$$

which corresponds in the time domain to

$$\ddot{e} + \omega_e \cdot \dot{e} + \omega_i \cdot \dot{e} + \omega_e \cdot \omega_i \cdot e = 0 \tag{114}$$

Solving Eq. (114) for the pseudo control $v_{\ddot{y}} = \ddot{y}$ results in

$$v_{\ddot{y},\text{des}} = \ddot{y}_{\text{des}} + \omega_e \cdot \dot{e} + \omega_i \cdot \dot{e} + \omega_e \cdot \omega_i \cdot e \tag{115}$$

which is inserted into the control law Eq. (111), resulting in

$$\begin{aligned} u_{\text{cmd}} &= \tfrac{1}{F_u(x,u)\cdot\omega_a} \cdot (\ddot{y}_{\text{des}} + \omega_e \cdot \dot{e} - F_x(x,u) \cdot \dot{x}) \\ &+ \tfrac{1}{F_u(x,u)\cdot\omega_a} \cdot (\omega_i \cdot \dot{e} + \omega_i \cdot \omega_e \cdot e) + u \end{aligned} \tag{116}$$

We consider the following two different choices of the desired inner-loop dynamics:

Case 1. desired inner-loop dynamics equal the true actuator dynamics, that is $\omega_i = \omega_a$.
Case 2. desired inner-loop dynamics correspond to scaled actuator dynamics, that is $\omega_i = \lambda \cdot \omega_a$.

Case 1. If we choose the desired inner-loop dynamics to equal the true actuator dynamics, that is $\omega_i = \omega_a$, then we obtain[57]

$$\begin{aligned} u_{\text{cmd}} &= \tfrac{1}{F_u(x,u)\cdot\omega_a} \cdot (\ddot{y}_{\text{des}} + \omega_e \cdot \dot{e} - F_x(x,u) \cdot \dot{x}) \\ &+ \tfrac{1}{F_u(x,u)} \cdot (\dot{e} + \omega_e \cdot e) + u \end{aligned} \tag{117}$$

We see that the contribution from the model dynamics and state derivatives, $F_x \cdot \dot{x}$, are inversely proportional to the actuator bandwidth ω_a. For infinitely fast actuators, that is infinite bandwidth ω_a the first term in Eq. (117) becomes zero and we obtain the classical INDI law, as shown in Note 12.

$$u_{\text{cmd}} = \frac{1}{F_u(x_0,u_0)} \cdot (v_{\text{des}} - \dot{y}_0) + u_0 \tag{118}$$

where $v_{\text{des}} = \dot{y}_{\text{des}} + \omega_e \cdot e$. That means that INDI corresponds to Eq. (117) under the assumption that ω_a is large enough such that the first part in Eq. (117) is negligible.

In case the actuator dynamics are not negligible, e.g., for actuator dynamics being not much faster than the system dynamics, then the first part in Eq. (117) can be re-introduced. The resulting closed-loop dynamics will correspond to the desired dynamics by re-introducing these terms.

Case 2. If we choose the desired inner-loop dynamics to equal the true actuator dynamics multiplied with a scalar, that is $\omega_i = \lambda \cdot \omega_a$, then we obtain[57]

$$\begin{aligned} u_{\text{cmd}} &= \tfrac{1}{F_u(x,u)\cdot\omega_a} \cdot (\ddot{y}_{\text{des}} + \omega_e \cdot \dot{e} - F_x(x,u) \cdot \dot{x}) \\ &+ \tfrac{1}{F_u(x,u)} \cdot \lambda \cdot (\dot{e} + \omega_e \cdot e) + u \end{aligned} \tag{119}$$

For infinitely fast actuators, that is, infinite bandwidth ω_a, the first term in Eq. (119) becomes zero. Then we obtain an INDI law, with scaling gain, as was proposed by[12]

$$u_{\text{cmd}} = \frac{1}{F_u(x_0,u_0)} \cdot \lambda \cdot (v_{\text{des}} - \dot{y}_0) + u_0 \tag{120}$$

where $v_{\text{des}} = \dot{y}_{\text{des}} + \omega_e \cdot e$. Ref. 12 suggested the input scaling to reduce the sensitivity to noise and to increase the stability margins and robustness with respect to time delays. Furthermore, Ref. 12 observed that a choice of $\lambda < 1$ reduces the closed-loop bandwidth. As Ref. 57 pointed out, the scaling gain can be interpreted as a modification of the desired inner-loop bandwidth. If the scaling gain is 1, then the desired inner-loop bandwidth corresponds to the true actuator bandwidth, and the conventional INDI control law is obtained. If $\lambda < 1$ then the desired inner-loop dynamics are slower than the actuator dynamics.

Note 12. To show that Eq. (118) indeed corresponds to the classical INDI law, we compare it to the INDI law that we obtain when using the Taylor approximation-based approach, discussed in Section 7, for the system in Eq. (107). The Taylor approximation of the output derivative $\dot{y} = F(x,u)$ is given by

$$\dot{y} \approx \dot{y}_0 + F_x(x_0,u_0) \cdot (x - x_0) + F_u(x_0,u_0) \cdot (u - u_0) \tag{121}$$

Solving for u by defining the pseudo control $v = \dot{y}$ results in

$$u_{\text{cmd}} = \frac{1}{F_u(x_0,u_0)} \cdot (v_{\text{des}} - \dot{y}_0 - F_x(x_0,u_0) \cdot \Delta x) + u_0 \tag{122}$$

By neglecting, as usual, the state-dependent term Δx, we obtain the same control law as in Eq. (118).

Note 13. Note that using $v_{\text{des}} = \dot{y}_{\text{des}} + \omega_e \cdot e$, we can reformulate the inversion law that we obtained using the Taylor approximation Eq. (122) as follows

$$\begin{aligned} u_{\text{cmd}} &= \tfrac{1}{F_u(x_0,u_0)} \cdot (\dot{e} + \omega_e \cdot e) \\ &- \tfrac{T_s}{F_u(x_0,u_0)} \cdot F_x(x_0,u_0) \cdot \dot{x}_0 + u_0 \end{aligned} \tag{123}$$

where $\dot{x}_0 = \frac{\Delta x}{T_s}$ with sample time T_s, is obtained by finite differences. We compare it to the ANDI control law given in Eq. (111), which we can formulate using Eq. (115), $v_{\ddot{y},\text{des}} = \ddot{y}_{\text{des}} + k_1 \cdot \dot{e} + k_0 \cdot e$, as follows

$$\begin{aligned} u_{\text{cmd}} &= \tfrac{1}{F_u(x_0,u_0)\cdot\omega_a} \cdot (\ddot{y}_{\text{des}} + k_1 \cdot \dot{e} + k_0 \cdot e) \\ &- \tfrac{1}{F_u(x_0,u_0)\cdot\omega_a} \cdot F_x(x_0,u_0) \cdot \dot{x}_0 + u_0 \end{aligned} \tag{124}$$

where $k_1 = \omega_e + \omega_i$ and $k_0 = \omega_e \cdot \omega_i$. If we choose $\omega_i = \omega_a$ it further simplifies to

$$\begin{aligned} u_{\text{c}} &= \tfrac{1}{F_u(x_0,u_0)} \cdot \left[\tfrac{1}{\omega_a} \cdot \ddot{y}_{\text{des}} + \left(\tfrac{\omega_e}{\omega_a} + 1\right) \cdot \dot{e} + \omega_e \cdot e\right] \\ &- \tfrac{1}{F_u(x_0,u_0)\cdot\omega_a} \cdot F_x(x_0,u_0) \cdot \dot{x}_0 + u_0 \end{aligned} \tag{125}$$

We see that the second part in Eq. (125) and Eq. (123) becomes equal if $\omega_a = \frac{1}{T_s}$. This corresponds to an actuator with time constant equal to the sample time. If T_s additionally approaches zero, then both laws become equal. This shows that for continuous time and infinitely fast actuators, the INDI law and the

ANDI law are the same for the given choice of desired error dynamics. In case these assumptions do not hold true, it shows what terms are actually neglected in INDI to make it an accurate inversion: The ANDI control law, in contrast to the INDI law, also considers the actuator bandwidth ω_a and a feedforward term consisting of the second derivative of y_{des}.

9.4. Overview of INDI-based controller variants

In this section, we provide an overview of the INDI-based controller variants introduced so far by summarizing their key characteristics in Table 1. This allows for a direct comparison between sensor-based, model-based, hybrid, and extended INDI (E-INDI) and actuator NDI (ANDI). The table contrasts the pseudo control types as well as the resulting control laws based on the following scalar SISO example system:

$$\begin{cases} \dot{x} &= F(x,u) \\ y &= x \end{cases} \tag{126}$$

The table concludes the derivation and discussion of the INDI-based controller variants. We see that the E-INDI and ANDI laws are on the $\dot{v}$ level and actively take the actuator dynamics into account via the actuator bandwidth ω_{OL} or ω_a. Note here that ANDI and E-INDI use a different notation for the actuator bandwidth. The pseudo control derivative $\dot{v}_{\text{des}}$ specifies in both cases the desired pseudo control dynamics or error dynamics. In E-INDI the state-dependent term $\frac{\partial F(x,u)}{\partial x} \cdot \dot{x}$ is usually neglected, which is a reasonable assumption for multicopter for example. For fixed-wing aircraft, this term might be significant. Then the state-dependent term can be included as a feedforward using a physical reference model or as feedback like in ANDI.

The key advantage of the E-INDI and ANDI approaches over conventional INDI is that the actuator and pseudo control loops are decoupled. This has the following benefits:

(1) The control effector dynamics can be accounted for in the inversion law;
(2) Multiple control effectors with different dynamics and bandwidths can be considered this way, addressing the challenge of pseudo control cross couplings introduced by the control effector dynamics (compare Note 8);
(3) For overactuated systems, the challenge that the pseudo control response in standard INDI is a control allocation (and potentially time) dependent combination of control effector dynamics can be addressed (compare Note 8);
(4) For each pseudo control channel, the desired dynamics can be specified independently, providing a straightforward way to design outerloops.

The following section will present an overview of the components that constitute INDI controllers.

10. Components of INDI controllers

INDI controller usually consist of several components, depicted in Fig. 10. Different approaches and common strategies can be found in the literature on how to design each of those components. Investigations of those strategies and approaches revealed specific challenges, and different solutions were developed. In the second part of this survey in Ref. 30, we focus on the components of INDI. For each component, we give an overview of the developments, discovered and discussed issues, and possible solutions that can be found in the literature. Here, we want to briefly summarize the different components of INDI as depicted in Fig. 10 and give an overview of their purpose.

The **reference model** together with an **error controller** generates the pseudo control command v_{des}. However, in several applications, outer control loops directly generate the reference commands, instead of using a reference model. The task of the **control allocation** is to allocate the desired pseudo control increments to effector command increments. For overactuated systems, additional constraints can be imposed. Furthermore, effector saturations and rate saturations might be taken into account. Several entities necessary in INDI control laws often need to be **estimated**. The control effectiveness, which might be needed in the control allocation module, might be, for example, estimated using an On-Board Plant Model (OBPM).

Furthermore, the current inputs u_0 and pseudo control v_0 have to be measured. If the corresponding sensors are unavailable or too expensive, as is the case in aircraft applications for angular accelerations, the signals have to be estimated. For this purpose, several approaches evolved to obtain estimates thereof and to deal with the side effects that might be introduced by the respective estimation techniques, e.g., applying synchronization.

Table 1 Overview of applications of INDI controllers.

INDI approach		Pseudo control $v = \dot{y}$	Control law
Sensor-Based (Section 4)		Measured $v = \dot{y}_0$	
Model-Based (Section 5)		Model-Based $v = \dot{y}_{\text{OBM}}$	
Hybrid (Section 6)	Linear blended	$v = (1 - K_{\text{aug}}) \cdot \dot{y}_{\text{OBM}} + K_{\text{aug}} \cdot \dot{y}_0$	$u = \left(\frac{\partial F(x,u)}{\partial u}\right)^{-1} \cdot (v_{\text{des}} - v) + u_0$
	Frequency weighted	$v = (1 - L(s)) \cdot \dot{y}_{\text{OBM}}(s) + L(s) \cdot \dot{y}_0(s)$ with filter $L(s)$ for instance a lowpass filter	
E-INDI (Section 9.1)		v can be model-based, sensor-based or hybrid. It is included in $\dot{v}_{\text{des}}$ and several variants are possible e.g.	$u = \frac{1}{\omega_{\text{OL}}} \cdot \left(\frac{\partial F(x,u)}{\partial u}\right)^{-1} \cdot \left(\dot{v}_{\text{des}} - \underbrace{\left(\frac{\partial F(x,u)}{\partial x}\right)^{-1} \cdot \dot{x}_{\text{ref}}}_{\text{optional}}\right) + u_0$
ANDI (Section 9.3)		$\dot{v}_{\text{des}} = \dot{v}_{\text{ref}} + K_{\dot{y}} \cdot (v_{\text{ref}} - v) + K_y \cdot (y_{\text{ref}} - y)$ $\dot{v}_{\text{des}} = \omega_{v,\text{des}} \cdot (v_{\text{des}} - v)$	$u = \frac{1}{\omega_a} \cdot \left(\frac{\partial F(x,u)}{\partial u}\right)^{-1} \cdot \left(\dot{v}_{\text{des}} - \left(\frac{\partial F(x,u)}{\partial x}\right)^{-1} \cdot \dot{x}\right) + u_0$

11. INDI and its relations to other control methods

In this section, we present a summary of the existing literature that compares Incremental Nonlinear Dynamic Inversion INDI with other control methods. Rather than conducting new comparisons or extending existing analyses, the aim here is to provide an overview of key findings from previous studies. This overview will help readers understand the context in which INDI has already been evaluated and how it compares to other advanced control methods. The control methods that INDI has been compared to, and which we summarize in this section, are the following ones:

(1) Proportional-Integral-(Derivative) PI(D) and Time-Delay Control (TDC), Section 11.1.
(2) Differential PI (DPI) and incremental DPI (iDPI), Section 11.2.
(3) Deadbeat Control, Section 11.3.
(4) Singular Perturbation Based Dynamic Inversion, Section 11.4.
(5) Adaptive Laws, Section 11.5

11.1. INDI and PI(D) and Time-Delay Control (TDC)

In 2017, Ref. 74 showed that INDI is equivalent to Time-Delay Control (TDC) - more familiar in the robotics community - and to PI(D) control. The following section will recap the results on the equivalence of INDI with PI(D) control, as it is the more common control methodology for various application domains. The INDI control law Eq. (20) for the single-state SISO system from Eq. (16), which has a relative degree of one, restated in the discrete domain, corresponds to

$$u[k] = g[k-1]^{-1} \cdot (\dot{e}[k-1] + \omega_e \cdot e[k-1]) + u[k-1] \tag{127}$$

where $v_{\text{des}}[k] = \dot{y}_{\text{ref}}[k] + \omega_e \cdot e[k]$, was used and where $e[k] = y_{\text{ref}}[k] - y[k]$ is the error and ω_e is a controller parameter. If the pseudo control cannot be directly measured, but only the system output, then the finite difference approximation provides

$$\dot{e}[k] = \frac{e[k] - e[k-1]}{T_s} \tag{128}$$

A PI control law can be formulated by

$$u(t) = K \cdot \left[e(t) + T_I^{-1} \cdot \int_0^t e(\tau) \mathrm{d}\tau \right] \tag{129}$$

with proportional gain K and integral gain $K \cdot T_I^{-1}$. The PI control law results in the discrete domain in

$$u[k] = K \cdot \left(e[k-1] + T_I^{-1} \cdot \sum_{i=0}^{k-1} T_s \cdot e[i] \right) \tag{130}$$

Subtracting two consecutive commands results in

$$\begin{aligned} &u[k] - u[k-1] = \\ &= K \cdot T_s \cdot \left(\tfrac{e[k-1]-e[k-2]}{T_s} + T_I^{-1} \cdot e[k-1] \right) \end{aligned} \tag{131}$$

Finally, using Eq. (128), the PI controller in incremental form is obtained

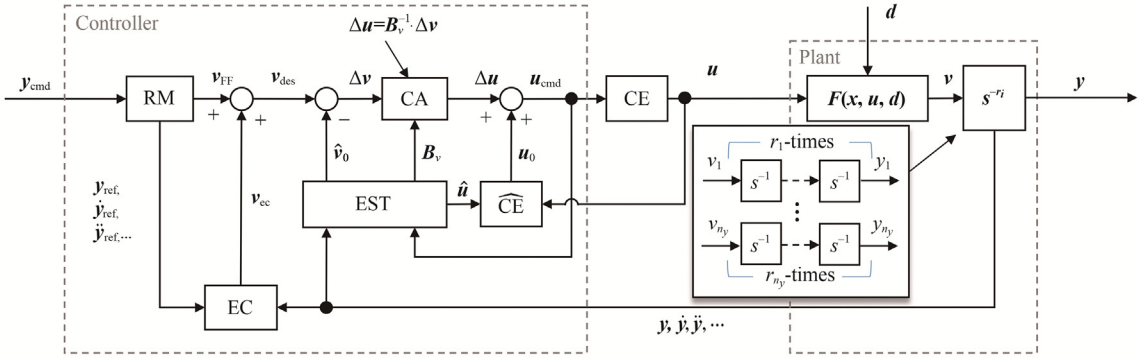
$$u[k] - u[k-1] = K \cdot T_s \cdot \left(\dot{e}[k-1] + T_I^{-1} \cdot e[k-1] \right) \tag{132}$$

which is identical to the INDI control law Eq. (127) if the gains are mapped according to $K \cdot T_s = g^{-1}[k-1]$ and $T_I^{-1} = \omega_e$. Notice that the incremental form of the PI controller introduces a finite difference of the error state in accordance with the INDI control law, which also uses the derivatives.

Hence, choosing the PI control gains as described above via INDI provides a more meaningful and systematic choice than heuristic methods: The gain value T_I^{-1} is determined by ω_e, which specifies the desired error dynamics, and the gain K contains the self scheduling properties via the control effectiveness g^{-1}. This result suggests how a PI can be tuned over INDI. It also suggests to only schedule K via the control effectiveness and not the whole set of PI controller gains over the flight envelope as usually done.

11.2. INDI, Differential PI (DPI) and incremental DPI (iDPI)

In 2023, Ref. 75 combined the classical Differential PI control with incremental control to iDPI. DPI control laws[76,77] are common practice in order to prevent in gain scheduled PI controllers the so-called hidden coupling terms or transients during mode switching. In classical flight control, it is a common practice to trim and linearize the nonlinear plant model at different envelope points, e.g., for multiple different combinations of static pressure (altitude) and dynamic pressure (velocity). At each trim point, a set of gains is calculated using the linear plant model. Later, these gains are interpolated



RM: Reference model, EC: Error controller, CA: Control allocation, EST: Estimation module,
CE: Control effector dynamcis, $\widehat{\text{CE}}$: Control effector estimator.

Fig. 10 Block diagram of a generic INDI controller with usual components.

as a function of the scheduling variables, e.g., altitude and velocity. This introduces additional feedback that might lead to oscillations or even a loss of stability, especially in case the scheduling variable is rapidly changing as the angle of attack, for example, such that a time scale separation between the scheduling variable and the controller bandwidth is not valid anymore. When linearizing the closed-loop system with the gain scheduled controller, this becomes visible as additional terms, the hidden coupling terms, appear that are not present in the linear design plant. DPI as proposed by Ref. 77, and for example used in the Eurofighter,[78–81] is a PI controller design where the linearization of the gain scheduled closed-loop system and the corresponding non-scheduled design closed-loop are identical. These DPI controllers avoid the so-called hidden coupling terms.

Ref. 75 presented the idea of combining DPI control laws with incremental control to iDPI. The resulting control law inherits the ability to avoid hidden coupling terms from DPI and the robustness characteristics from INDI.

Fig. 11 shows an example of a DPI controller, and Fig. 12 an iDPI controller, where the controller gains are scheduled and change over time. Comparing these figures, we can see how closely they are related. We see that we can reformulate the incremental loop of INDI such that they are identical except for the effective actuator dynamics. Ref. 75 showed that the iDPI controller has an increased robust performance with respect to variations in most of the model parameters and an enhanced disturbance rejection. Additionally, it is more robust against time delays at the input.

Besides the analytical considerations, they perform simulations on a linear example system. Furthermore, they confirm the results by retrofitting an industry-level PI baseline controller[82–84] to the iDPI control law[75] for the longitudinal motion of a fixed-wing aircraft using a high-fidelity simulation model.

Finally, they reformulate the iDPI structure to match the concept of model following INDI with synchronization on the actuator feedback path as depicted in Fig. 13 where the reference signals come from a physical model. Physical reference models have been used both with NDI and INDI.[85–87] We address this topic in more detail in the second part of this survey in Ref. 30.

Comparing the derivatives rather than the total states avoids that in steady state conditions the errors in the different variables, due to uncertainties, are fighting against each other. This problem can be avoided using the derivatives since their steady-state value is zero, providing a simpler and more straightforward solution than the approach proposed in Ref. 85. Ref. 85 addressed this problem by a blending strategy in the error controller.

The incremental Differential PI (iDPI) controller has also been applied to control the lateral motion of a fixed-wing aircraft and has been successfully flight tested in Ref. 88.

11.3. INDI and deadbeat control

In 2022, Pfeifle and Fichter,[89] apply discrete deadbeat control theory to the minimum settling time angular acceleration control problem. The resulting control law corresponds to an INDI control law with an additional gain as depicted in Fig. 14.

In Fig. 14, $A(z)$ denotes the discrete actuator dynamics and B the control effectiveness. The gain depends on the sampling time T_s and actuator time constant τ as follows

$$K_{\mathrm{DB}} = \frac{1}{1 - \mathrm{e}^{-\frac{T_s}{\tau}}}. \tag{133}$$

This gain corresponds according to Ref. 89 to the gain proposed in Ref. 65 to overdrive the actuators. However, a common issue in deadbeat control is its tendency to generate input commands with large magnitude, especially for small ratios: $\frac{T_s}{\tau} \to 0$. This case[89] proposes to use Dahlin's controller where the closed-loop transfer function can be specifically specified to be first-order and with a steady-state gain of 1. For first-order actuators, the parameters of this controller can be chosen such that the controller reduces to the standard INDI controller.

11.4. INDI and singular perturbation based dynamic inversion

Hovakimyan and Lavretsky[90,91] introduce a Dynamic Inversion law derived from time scale separation and singular perturbation theory to deal with input non-affine systems of the form $\dot{x} = f(x,u)$, which cannot be explicitly solved for u. As we show in the following, their dynamic inversion law shows similarities to INDI under certain conditions. They introduce the following dynamic system

$$\epsilon \cdot \dot{u} = -\mathrm{sign}\left(\frac{\partial f}{\partial u}\right) \cdot (f(x,u) - \dot{x}_{\mathrm{des}}) \tag{134}$$

for which $f(x,u) \to \dot{x}_{\mathrm{des}}$ if $\epsilon \to 0$. The dynamics in Eq. (134) can be reformulated as

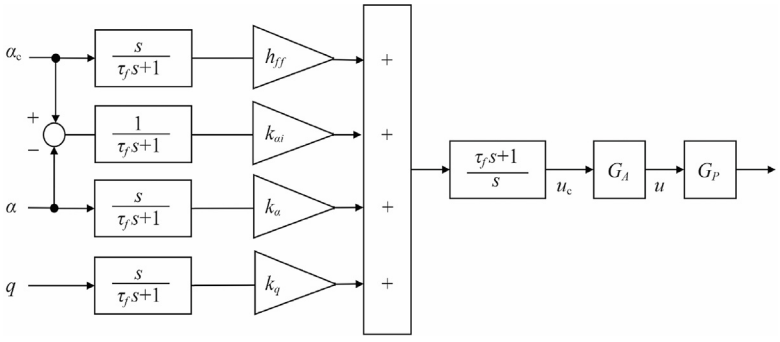


Fig. 11 Classical Differential Gain Scheduled PI Controller (DPI) adapted from Ref. 75.

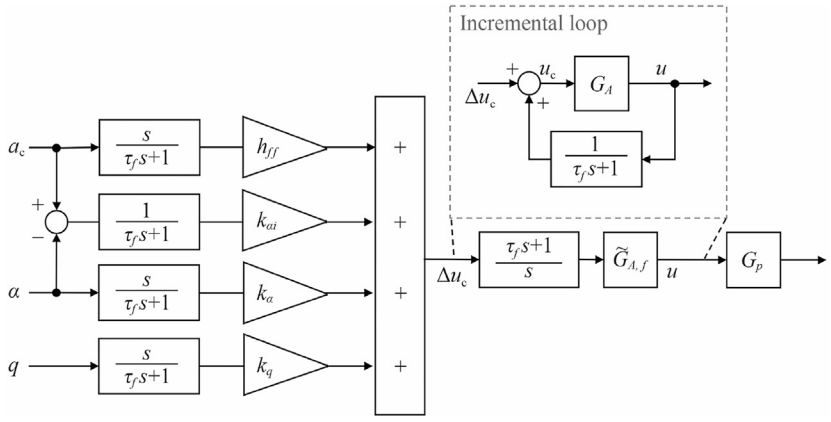


Fig. 12 Incremental Differential Gain Scheduled PI controller (iDPI) adapted from Ref. 75.

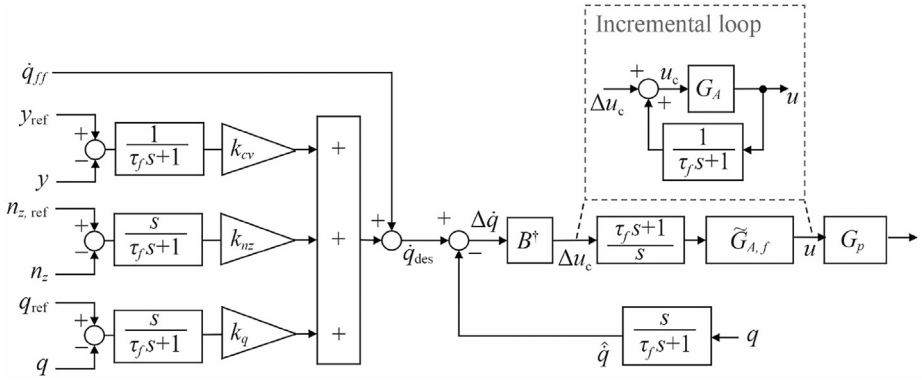


Fig. 13 Model following iDPI adapted from Ref. 75.

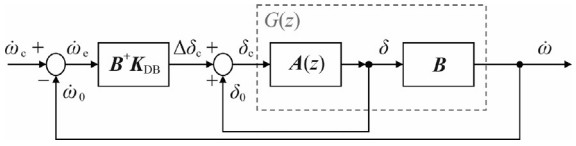


Fig. 14 Deadbeat controller with K_{DB} reformulated as INDI inner-loop adapted from Ref. 89.

$$\dot{u} = \frac{1}{\epsilon} \cdot \text{sign}\left(\frac{\partial f}{\partial u}\right) \cdot (\dot{x}_{\text{des}} - \dot{x}_{\text{mdl}}) \tag{135}$$

where $\dot{x}_{\text{mdl}} = f(x,u)$ is a model-based estimate of the state derivative. If u is now calculated via integration of Eq. (135) with euler backward, we obtain

$$\begin{aligned} u &= \dot{u} \cdot T_s + u_0 \\ &= \tfrac{1}{\epsilon} \cdot T_s \cdot \text{sign}\left(\tfrac{\partial f}{\partial u}\right) \cdot (\dot{x}_{\text{des}} - \dot{x}_{\text{mdl}}) + u_0 \end{aligned} \tag{136}$$

where u_0 is the input from the previous time step. In case there is a difference in the desired system response and the actual one, the control law in Eq. (136) will increase the control effector command in the direction, which causes the desired reaction. The slope or step size is adjusted by $\frac{T_s}{\epsilon}$. That means the smaller ϵ, the higher the gain. If ϵ is chosen as $|\frac{\partial f}{\partial u}| \cdot T_s = |B_v| \cdot T_s$ the control law Eq. (136) will result in the model-based INDI law Eq. (23). Note, that this comparison adresses SISO systems.

11.5. INDI and adaptive laws

Since the RESTORE project,[33,92,93] the combination of NDI with adaptive control, such as Model Reference Adaptive Control (MRAC), was thoroughly investigated.[27,28,87,94–101] For a pseudo control given by

$$\boldsymbol{v} = \boldsymbol{a}_v(\boldsymbol{x}) + \boldsymbol{B}_v(\boldsymbol{x}) \cdot \boldsymbol{u} \tag{137}$$

with NDI law

$$\boldsymbol{u} = \boldsymbol{B}_v^{-1}(\boldsymbol{x}) \cdot (\boldsymbol{v} - \boldsymbol{a}_v(\boldsymbol{x})) \tag{138}$$

the adaptation would address the uncertain model information given in $\boldsymbol{a}_v$ and $\boldsymbol{B}_v$. As discussed in Section 4, the model information given by $\boldsymbol{a}_v$ is replaced in INDI by measurements. Hence, only the necessity to adapt for the model information given by $\boldsymbol{B}_v$remains.

Several publications address the adaptation of the control effectiveness $\boldsymbol{B}_v$ in the context of INDI and other modern control techniques. Adapting the control effectiveness in the context of control allocation is often called adaptive control allocation. Typically, one can divide the approaches into direct and indirect adaptive control allocations similar to MRAC. In direct adaptive control allocation, the adaptation occurs in the

pseudo control space and does not directly estimate the control effectiveness.[102–104] The indirect approaches aim to either estimate the full control effectiveness matrix $\boldsymbol{B}_v$[105,106] or just one parameter per effector as the control effectiveness vector $\boldsymbol{\lambda}$.[107–109] The approaches rely on MRAC, online parameter estimation, and concurrent learning MRAC using gradient descent,[107] recursive least squares,[105,106,108] or Kalman filtering techniques.[109] All of those approaches achieve fault tolerance and adaptation to model uncertainties.

With the adaptation might come the necessity to recompute the pseudoinverse of the control effectiveness matrix when respective control allocation is used. The computation using the Singular Value Decomposition (SVD) poses a challenge in safety–critical systems due to its significant computational workload. Ref. 110 proposed a numerically efficient, deterministic, and transparent rank-one update-based algorithm for control allocation with changing control effectiveness matrices.

12. INDI in different types of control loops

This section gives an overview of some typical applications of INDI. Note that this list is not complete. Neither all applications are captured, nor are all publications addressing this application listed for the selected applications. Here, we intend to give a rough overview, stressing that besides inner-loop control, INDI has also been successfully applied to other control loops. For example, already back in 1995, Ref. 24 described an outer-loop control law that calculates a pitch rate command based on the desired increment in vertical body acceleration,[24] which conceptually resembles INDI. As described in Section 3, Smith, three years later, was the first to introduce an INDI like inversion for the pitch inner-loop control in Ref. 35.

12.1. Inner-loop for rotational rate control

In flight control applications, INDI is most commonly applied to control the rotational dynamics as done, for example, in Refs. 9,52,53, including an attitude loop (Euler angles or quaternions) wrapped around it. As inverting the rotational dynamics might require knowledge of the aerodynamic model in many cases, INDI can replace this susceptibility to model errors by measurements of the control effector inputs u_0 and angular accelerations $\dot{\boldsymbol{\omega}}_0 = \boldsymbol{\nu}$. The resulting angular rate controller is the most inner control loop, generating the actuator commands according to

$$\boldsymbol{u} = \boldsymbol{B}_{\boldsymbol{\nu}}^{-1} \cdot (\boldsymbol{\nu}_{\text{des}} - \dot{\boldsymbol{\omega}}_0) + \boldsymbol{u}_0 \tag{139}$$

Different approaches to obtaining and calculating the pseudo control $\boldsymbol{\nu} = \dot{\boldsymbol{\omega}}_0$ evolved in case it can not be directly measured. The second part of this survey in Ref. 30, where we address the different components in INDI, discusses these approaches. Besides the most common application as an inner-loop control law, INDI has also been successfully used in other control loops, as the following subsections show.

12.2. Cascaded INDI for path following via aerodynamic attitudes

In 2021, Ref. 50 presented a cascaded INDI controller, similar in structure to the one presented in Ref.111 and depicted in Fig. 15. As visible in Fig. 15, an outer INDI loop is introduced, besides the common INDI inner-loop to control the rotational dynamics. A proportional rate loop generates the angular acceleration commands of the INDI inner-loop. The rate commands are obtained by a NDI law inverting the relationship between the rates and aerodynamic attitude derivatives $\left[\dot{\mu}, \dot{\alpha}, \dot{\beta}\right]^{\mathrm{T}}$, which is a purely kinematic relationship. Hence, no model uncertainties are present.

The commanded $\left[\dot{\mu}_{\mathrm{c}}, \dot{\alpha}_{\mathrm{c}}, \dot{\beta}_{\mathrm{c}}\right]^{\mathrm{T}}$ are calculated similarly to the rate loop from proportional feedback of the error between the commanded and measured attitudes. The attitude commands $[\mu_{\mathrm{c}}, \alpha_{\mathrm{c}}]^{\mathrm{T}}$ are finally obtained from an INDI tracking control loop, while the sideslip angle command is $\beta_{\mathrm{c}} = 0$. The INDI tracking control loop calculates $[\mu_{\mathrm{c}}, \alpha_{\mathrm{c}}]^{\mathrm{T}}$ and propeller speed Ω_{c}, using the measurements $[\mu_0, \alpha_0, \Omega_0]^{\mathrm{T}} (= u_0)$ and the measurements $\left[\dot{V}_{a,0}, \dot{\chi}_0, \dot{\gamma}_0\right] (= \boldsymbol{\nu}_0)$. The commands $\left[\dot{V}_{a,\mathrm{c}}, \dot{\chi}_{\mathrm{c}}, \dot{\gamma}_{\mathrm{c}}\right] (= \boldsymbol{\nu}_{\text{des}})$ are again obtained from proportional feedback of the error in $[V_a, \chi, \gamma]$ and the respective commands $[V_{a,\mathrm{c}}, \chi_{\mathrm{c}}, \gamma_{\mathrm{c}}]$ are generated by another NDI position controller. This cascaded INDI framework provided good wind and gust rejection properties due to the incremental tracking control loop and because aerodynamic attitude is controlled instead of Euler angles.

12.3. Cascaded INDI for path following via Euler angles

Ref. 52 presented separate control laws for the fixed-wing and hover flight for a quad-plane. For fixed-wing flight, the inner-loop is the common INDI law given by Eq. (139) where u is the surface deflections. In hover, the input u_h are the lift motor rotational speeds $(\Omega_1, \Omega_2, \Omega_3, \Omega_4)$ and are allocated based on the increments in angular and vertical accelerations based on INDI as follows

$$u_h = B_{\nu,h} \cdot \left(\nu_h - \begin{bmatrix} \dot{\omega}_0 \\ \ddot{h}_0 \end{bmatrix} \right) + u_{h,0} \tag{140}$$

The pseudo control signal for the rotational accelerations is in both cases, calculated based on the reference signals from a second-order reference model in Euler angles $\Phi_{\text{ref}}, \Theta_{\text{ref}}$ and their derivatives up to $\ddot{\Phi}_{\text{ref}}, \ddot{\Theta}_{\text{ref}}$ and the respective measurements. For the heading, the first and second derivatives are used only and obtained from a first-order reference model. The error controller structures, which calculate ν and ν_h, slightly differ for hover and fixed-wing. The relation between rates and Euler angles over the strap-down equation is neglected in both cases. The pseudo control signal for the $\ddot{h}$ in hover is obtained in a similar manner with again slightly differing error control law structure.

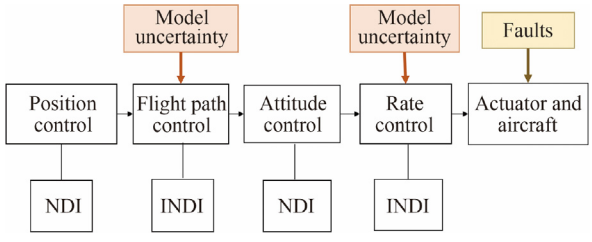


Fig. 15 Cascaded INDI adapted from Ref. 111 and Ref. 50.

The INDI-based outer-loop for fixed-wing flight calculates the forward thrust F_p command and pitch angle command by

$$u_{\mathrm{OL}} = \begin{bmatrix} F_{p,\mathrm{cmd}} \\ \Theta_{\mathrm{cmd}} \end{bmatrix} = B_{v,\mathrm{OL}}^{-1} \cdot \left(v_{\mathrm{OL}} - \begin{bmatrix} \dot{V}_{a,0} \\ \dot{\gamma}_0 \end{bmatrix} \right) + u_{\mathrm{OL},0} \tag{141}$$

with measured airspeed derivative $\dot{V}_{a,0}$ and flight path angle derivative $\dot{\gamma}_0$. Linear controllers are used to generate the pseudo control commands given by v_{OL} based on the error in $V_{a,\mathrm{cmd}} - V_{a,0}$ and $\gamma_{\mathrm{cmd}} - \gamma_0$. Here γ_{cmd} is obtained from

$$\gamma_{\mathrm{cmd}} = \frac{\dot{h}_{\mathrm{cmd}}}{V_a} = \frac{K_h}{V_a} \cdot (h_{\mathrm{cmd}} - h) \tag{142}$$

A linear error controller in course angle χ provides the bank angle command Φ_{cmd}. An error controller in β calculates the command $\dot{\Psi}_{\mathrm{cmd}}$. Fig. 16 depicts the complete controller structure for fixed-wing flight.

In hover flight the Euler angle commands Φ_{cmd} and Θ_{cmd} denoted as $u_{\mathrm{OL,h}}$ are obtained by the INDI law

$$u_{\mathrm{OL,h}} = B_{v,\mathrm{OL,h}} \cdot \left(v_{\mathrm{OL,h}} - \begin{bmatrix} \dot{u}_0 \\ \dot{v}_0 \end{bmatrix} \right) + u_{\mathrm{OL,h},0} \tag{143}$$

where $\dot{u}$ and $\dot{v}$ are the longitudinal and lateral accelerations in the body fixed frame. Linear error controllers in u and v provide the pseudo control commands. An error controller in χ calculates $\dot{\Psi}_{\mathrm{cmd}}$. Fig. 17 depicts the control structure in hover flight.

13. INDI for different vehicle types

INDI has been successfully applied to a variety of different vehicle types. The reason is that INDI is able to perform decoupling of the axes and requires less model information than NDI. Both properties are valuable when considering novel aerial vehicles, which often have coupled dynamics. Furthermore, their aerodynamics are often not well known, and novel complex effects that influence the dynamics cannot yet be modeled accurately due to a lack of experience and knowledge. Note that the following lists do not capture all relevant INDI publications for the respective vehicle types. We intend to give an overview of the large variety of vehicle types INDI has been successfully applied to.

For a more in-depth analysis of each component of INDI, including its implementation across different platforms and a detailed comparison of trends and specific adaptations for various vehicles, please refer to our second paper titled "Advancements in Incremental Nonlinear Dynamic Inversion and its Components: A Survey on INDI – Part II".[30] Before diving into the different subsections, Table 2 provides a summary of vehicle types where INDI was applied and whether flight tests were conducted.

13.1. High performance aircraft

In 2000/2001 Bacon and Ostroff demonstrate their incremental law (see Section 7) via simulations on the Innovative Control Effectors (ICE) Tailless Fighter Aircraft Configuration 101–3,[33,56] the same that also Buffington uses in Ref. 26. They recognized that "...the actuator loop is effectively an integrator created by feeding back the actuator position",[33] which could lead to wind-up and driving actuators into saturation because of accumulating uncertainties. They proposed adding a first-order reference model for each actuator to avoid this. They determine the feedback of each control input for the inversion law, as the minimum of the absolute values of the model and measurement. When choosing the measurement, they reset the model to the current measurement value. Such onboard models are used for all actuators in Bacon and Ostroff's paper.[33]

In 2000, Smith and Berry[10] flight tested his inversion control law (see Section 3) in the pitch channel on the Vectored thrust Aircraft Advanced flight Control (VAAC) Harrier. The VAAC Harrier is able to operate at high speeds but also can perform VTOL using a Reaction Control System. In 2002, Bordignon and Bessolo incremental control law[41] - a model-based INDI (Section 5), where the current pseudo control is calculated using a high fidelity onboard model of aerodynamics and propulsion - is demonstrated successfully in real flight on Short Take-Off and Vertical Landing (STOVL) variant of Lockheed Martin's Joint Strike Fighter (the X-35B).

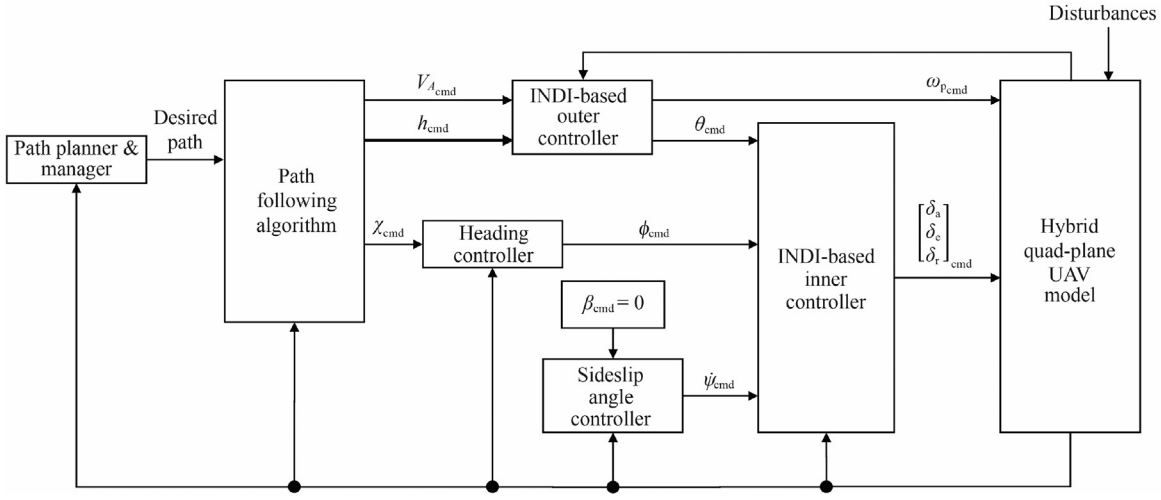


Fig. 16 Fixed wing controller adapted from Ref. 52.

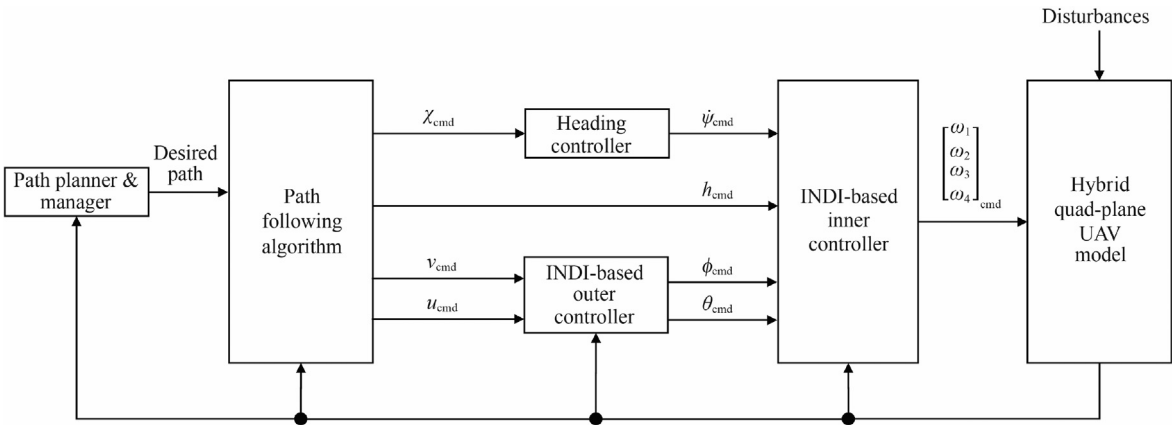


Fig. 17 Hover controller adapted from Ref. 52.

Table 2 Overview of applications of INDI controllers.

Plant	Simulation-only	Flight test
Fighter	Refs. 26,33,53,56,85,112	Refs. 10,41,42
Fixed wing	Ref. 113	Refs. 7–9,114,115
Helicopter	Refs. 36,116,117	
Multicopter	Ref. 63	Refs. 2–4,118,119
Tilt-rotor	Refs. 48,51,54,55,120	Refs. 5,121
Tilt-wing & Tailsitter	Refs. 58,60,122	Refs. 6,123,124
Quadplane	Ref. 52	Refs. 125
Cyclocopter		Ref. 126
Airship	Refs. 11,12,127	
Other	Refs. 13–18	

In 2019, Ref. 53 applied the common INDI inner-loop control law Eq. (139) with the angular accelerations in stability frame to a F-16 model. Ref. 53 focused on the control allocation, which uses a redistributed scaled pseudo inverse. An OBPM updates the effectiveness matrix $\boldsymbol{B}_v$, which calculates the Jacobian matrices using numerical perturbation.

Ref. 85 controlled the longitudinal channel of a piloted fixed-wing, all-attitude, high-performance aircraft using INDI. The following features, which aim to realize carefree handling for the pilot over the entire flight envelope, can be realized by combining the INDI law with an incremental reference model:

(1) Model following approach: an incremental physical reference model is used to shape the desired dynamics to satisfy the desired handling qualities. The feedback path can be tuned with regard to disturbance rejection, stability, and robustness.
(2) Control variable blending: the proposed control law structure and reference model structure allow the blending of the control variable over different areas of the flight envelope. This change in the control variable supports the tasks and limitations in the different flight envelope domains.
(3) Flight envelope protections: the chosen structure allows the incorporation of flight envelope protections such as the phase plane-based protections discussed in Refs. 128,129.
(4) Gravity rate compensation, kinematic coupling compensation, pilot relief functions, turn coordination, and turn compensation: These functions are incorporated to support the pilot in challenging tracking tasks (e.g., formation flying or precision approaches).

13.2. Fixed-wing aircraft

In 2005, Dickman[7] performs successful flight tests with Smith's incremental inversion law (from Ref. 10) on a 30 mm wingspan, fixed-wing Micro Aerial Vehicle (MAV). He describes the usage of rotational accelerations, being calculated through complementary filtering.

In 2018, Ref. 8 the Netherlands Aerospace Center and German Aerospace Center, together with TU Delft have applied INDI to the Cessna Citation II flying research platform (PH-LAB), marking the first application of INDI to a CS-25 aircraft. Subsequentially, Ref. 71 extended the work by Ref. 8 by an incremental current controller of the electro-mechanical actuators. The new actuator control strategy was successfully tested in flight using the PH-LAB aircraft. Besides this, INDI has been successfully applied to various fixed-wing model airplanes such as in Ref. 9, which provides flight test results.

13.3. Helicopter

In 2013, Ref. 116 applied the INDI methodology to helicopter flight control for a single main rotor and tail rotor helicopter and evaluated it in simulation. In Ref. 36, an INDI controller

is presented for the DLR ACT/FHS research helicopter, which is a modified EC135 helicopter equipped with a fly-by-wire/fly-by-light control system. The approach is denoted Incremental Dynamic Inversion (IDI), as linear models are considered. Hence, the word 'nonlinear' in INDI is omitted. Pilot tests are performed in a high-fidelity flight simulator, and Level 1 Handling Qualities (HQ) for the Mission Task Elements Pirouette and Slalom according to ADS-33E are attested. To account for actuator rate saturations, Ref. 36 evaluated different methods.

13.4. Multicopter UAV

INDI was applied successfully to multicopters as for example demonstrated in Refs. 2–4. In Ref. 63 INDI is applied to a multicopter mini drone. In 2015, Ref. 3 demonstrated INDI in real flight test on a micro aerial vehicle. The complete algorithm comprises:

(1) an adaptive part, where the control effectiveness is updated online using a least mean squares algorithm,
(2) The incorporation of the momentum of the propellers in the controller - since the angular accelerations of their control effectors (rotors of the MAV) have a direct impact on the rotational acceleration of the aircraft as well (opposing torque moment), the angular acceleration actually is a function of control effector state u and control effector state derivative $\dot{u}$.
(3) a synchronization technique to accurately manage delays that occur when calculating angular accelerations from angular rate measurements using low-pass filters and differentiation.

13.5. Tilt-rotor and tilt-wing and tailsitter aircraft

Examples for INDI control laws applied to tilt-rotor aircraft can be, for example, found in Refs. 5,120,121. Ref. 5 proposed a unified INDI controller that can handle all flight phases of a Vertical Take-Off and Landing (VTOL) transition vehicle.

The system dynamics in the case of tilt rotors are input non-affine. INDI can be beneficial here as it locally linearizes the control effectiveness such that an input affine formulation approximates the system. Ref. 54 used virtual controls corresponding to polar coordinates of the overall specific thrust vector in the body-fixed frame and suggest a cascaded controller with a INDI inner-loop and outer-loop. Ref. 48 used virtual controls, too, together with a velocity-dependent blending of the rotor and surface controls in transition. Ref. 51 developed an INDI controller for position and attitude control for a tilt rotorcraft.

In 2022, Ref. 124 applied INDI to a tailsitter and evaluate it successfully in flight tests. Including a jerk reference through a feedforward term improves the tracking of trajectories with fast-changing accelerations. Ref. 6 used INDI for attitude control for a tailsitter micro air vehicle with tilt rotors. The attitude outer-loop calculates rate commands based on quaternion error feedback. Thrust vectoring is used exclusively for control moment generation. The tilt angles and angular velocities of the motors are the control inputs. Indoor and outdoor flight tests are conducted. Further, Refs. 58,60,122 and Ref. 123 present different INDI based control strategies for tilt-wing aircraft.

13.6. Quad-plane UAV

Ref. 52 used INDI to control a quad-plane. They developed different control structures to control hover and fixed-wing flight. Both the inner-loop and the outer-loop used in both cases (hover and fixed-wing controller) INDI. Ref. 125 developed an INDI controller for a variable skew quad plane.

13.7. Cyclocopter UAV

Ref. 126 applies an INDI control law to a cyclocopter and present flight test results. The cyclocopter is based on a novel propulsion system, a cyclogyro rotor concept, and an improved variant of the Voith-Schneider propeller.

13.8. Airship

Ref. 11 applied INDI to a linear airship model to control the lateral motion. The presented simulation results include wind disturbances and a path following loop. In 2022, Ref. 12 applied INDI to control the lateral motion of an airship. They present simulation results based on a nonlinear model that includes vehicle aerodynamics, actuators, and sensors. In 2024, Ref. 127, presented an INDI controller for the velocity/attitude control of a six-propeller electrical airship with independent tilting propellers, enabling flexible maneuverability.

13.9. Other

Besides aircraft, INDI has been applied to various other systems. Ref. 13 applies an INDI for example to an unmanned underwater vehicle. In Refs. 14,15 INDI is applied to spacecraft and in Ref. 16 to pressurized water reactor based small modular reactors. Ref. 17 used INDI to control a hydraulic hexapod flight simulator motion system, and Ref. 18 to control a pneumatic actuation system.

14. Conclusions

The presented survey study highlights the main methodologies related to INDI. The main streams, being sensor-based INDI, model-based INDI, and hybrid INDI, are derived and related to each other. Model-based INDI and NDI are equivalent for input affine systems with an equal number of inputs and outputs, where the control effectiveness is invertible. For the other cases, the differences between both approaches are discussed.

The hybrid INDI approaches combine the sensor- and model-based parts. One possibility is to combine them with a proportional blending that might be scheduled over the envelope and allows adaptation to varying model uncertainties. Other hybrid approaches are based on the frequency domain and combine the model and sensor-based parts with complementary filters, such as highpass and lowpass. This allows for noise attenuation.

Further advances to INDI, such as the E-INDI and ANDI, which consider actuator dynamics, are discussed in detail. Their main advantage is that they allow the specification of desired dynamics for each pseudo control channel separately, also in case the system is MIMO and multiple actuators with

different bandwidths are present. Several publications relate INDI to other control concepts. Selected comparisons are summarized here, setting INDI into context to other control methods. Finally, applications of INDI are summarized with respect to control loops for which it has been used and with regard to different types of vehicles and plants. A table details the different application and system types and highlights simulation-based investigations, and studies with real flight tests.

This survey focuses mainly on the methodologies of INDI, how and why they evolved and the developed advances. However, INDI controllers comprise more components such as control allocation, reference models, error controllers, state and pseudo control estimation, and many more. These components underly specific challenges, and extensive research has focused on solutions to these challenges. The second part of this two-part series of surveys in INDI focuses on these components and their advancements.

CRediT authorship contribution statement

Agnes Christine Steinert: Writing – review & editing, Writing – original draft, Conceptualization.**Stefan Raab**: Writing – review & editing, Writing – original draft, Visualization, Validation.**Florian Holzapfel**: Writing – review & editing, Supervision, Funding acquisition.**Haichao Hong**: Writing – review & editing.**Simon Hafner**: Writing – review & editing, Writing – original draft, Visualization.

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References

1. Rupprecht TA, Steinert A, Kotitschke C, et al. INDI control law structure for a MEDEVAC eVTOL and its reference models: Feedforward, physical limitations, and innerloop dynamics for optimal tracking. Reston: AIAA; 2024. Report No.: AIAA-2024-4425.
2. Akkinapalli VS, Holzapfel F. Incremental dynamic inversion based velocity tracking controller for a multicopter system. Reston: AIAA; 2018. Report No.: AIAA-2024-1345.
3. Smeur EJJ, Chu Q, de Croon GCHE. Adaptive incremental nonlinear dynamic inversion for attitude control of micro air vehicles. *J Guid Contr Dyn* 2015;**39**(3):450–61.
4. Wang ZX, Zhao J, Cai ZH, et al. Onboard actuator model-based incremental nonlinear dynamic inversion for quadrotor attitude control: Method and application. *Chin J Aeronaut* 2021;**34**(11):216–27.
5. Raab SA, Zhang JN, Bhardwaj P, et al. Proposal of a unified control strategy for Vertical take-off and landing transition aircraft configurations. Reston: AIAA; 2018. Report No.: AIAA-2018-3478.
6. Lovell-Prescod GHLH, Ma ZQ, Smeur EJJ. Attitude control of a tilt-rotor tailsitter micro air vehicle using incremental control. *2023 International conference on unmanned aircraft systems (ICUAS)*. Piscataway: IEEE Press; 2023.
7. Dickman G. Rapid prototyping of micro air vehicle control systems. Reston: AIAA; 2005. Report No.: AIAA-2005-7068.
8. Grondman F, Looye G, Kuchar RO, et al. Design and flight testing of incremental nonlinear dynamic inversion-based control laws for a passenger aircraft. Reston: AIAA; 2018. Report No.: AIAA-2018-0385.
9. Xin HB, Chen QY, Wang P, et al. The control performance analysis of the incremental nonlinear dynamic inverse method and flight test. *Proceedings of 2022 international conference on autonomous unmanned systems (ICAUS 2022)*. Singapore: Springer Nature Singapore; 2023.
10. Smith P, Berry A. Flight test experience of a nonlinear dynamic inversion control law on the VAAC harrier. Reston AIAA; 2000. Report No.: AIAA-2000-3914.
11. Azinheira JR, Moutinho A, Carvalho JR. Lateral control of airship with uncertain dynamics using incremental nonlinear dynamics inversion. *IFAC-PapersOnLine* 2015;**48**(19):69–74.
12. Cordeiro RA, Marton AS, Azinheira JR, et al. Increased robustness to delay in incremental controllers using input scaling gain. *IEEE Trans Aerosp Electron Syst* 2022;**58**(2):1199–210.
13. Chen GM, Liu A, Hu JH, et al. Attitude and altitude control of unmanned aerial-underwater vehicle based on incremental nonlinear dynamic inversion. *IEEE Access* 2020;**8**:156129–38.
14. Acquatella P, Van Kampen EJ, Chu QP. A sampled-data form of incremental nonlinear dynamic inversion for spacecraft attitude control. Reston: AIAA; 2022. Report No.: AIAA-2022-0761.
15. Acquatella BP, Chu QP. Agile spacecraft attitude control: an incremental nonlinear dynamic inversion approach. *IFAC-PapersOnLine* 2020;**53**(2):5709–16.
16. Bose D, Hazra A, Mukhopadhyay S, et al. A co-ordinated control methodology for rapid load-following operation of a pressurized water reactor based small modular reactor. *Nucl Eng Des* 2020;**367**:110748.
17. Huang YZ, Pool DM, Stroosma O, et al. Incremental nonlinear dynamic inversion control for hydraulic hexapod flight simulator motion systems. *IFAC-PapersOnLine* 2017;**50**(1):4294–9.
18. Das H, Pool D, van Kampen EJ. Incremental nonlinear dynamic inversion control of long-stroke pneumatic actuators. *2021 European control conference (ECC)*. Piscataway: IEEE Press; 2021.
19. Falb P, Wolovich W. Decoupling in the design and synthesis of multivariable control systems. *IEEE Trans Autom Contr* 1967;**12**(6):651–9.
20. Nazar S, Rekasius Z. Decoupling of a class of nonlinear systems. *IEEE Trans Autom Contr* 1971;**16**(3):257–60.
21. Asseo SJ. Decoupling of a class of nonlinear systems and its application to an aircraft control problem. *J Aircr* 1973;**10**(12):739–47.
22. Bugajski D, Enns D, Elgersma M. A dynamic inversion based control law with application to the high angle-of-attack research vehicle. Reston: AIAA; 1990. Report No.: AIAA-1990-3407.

23. Snell SA, Enns DF, Garrard Jr WL. Nonlinear inversion flight control for a supermaneuverable aircraft. *J Guid Contr Dyn* 1992;**15**(4):976–84.
24. Smith P, Patel Y. Translational motion control of VSTOL aircraft using nonlinear dynamic inversion. Reston: AIAA; 1995. Report No.: AIAA-1995-3452.
25. Colgren R, Enns D, Colgren R, et al. Dynamic inversion applied to the F-117A. Reston: AIAA; 1997. Report No.: AIAA-1997-3786.
26. Buffington JM. Modular control law design for the innovative control effectors (ICE) tailless fighter aircraft configuration 101–3. Wright-Patterson AFB: Storming Media; 1999. Report No. AFRLVA-WP-TR-1999-3057.
27. Wise KA, Brinker JS, Calise AJ, et al. Direct adaptive reconfigurable flight control for a tailless advanced fighter aircraft. *Int J Robust Nonlinear Contr* 1999;**9**(14):999–1012.
28. Holzapfel F. Nonlinear adaptive control of unmanned aerial vehicles [dissertation]. Munich: Technische Universität München, 2004 [German].
29. Dan CN. F-35 high angle of attack flight control development and flight test results. Reston: AIAA; 2019. Report No.: AIAA-2019-3227.
30. Steinert A, Raab S, Hafner S, et al. Advancements in incremental nonlinear dynamic inversion and its components: A survey on INDI – Part II. *Chin J Aeronaut* 2025;**38**(11):103591.
31. Slotine JJE, Li W. *Applied nonlinear control*. Englewood Cliffs: Prentice Hall; 1991. p. 216–36.
32. Khalil HK. *Control of nonlinear systems*. Englewood Cliffs: Prentice Hall; 2002. p. 505–50.
33. Bacon B, Ostroff A. Reconfigurable flight control using nonlinear dynamic inversion with a special accelerometer implementation. Reston: AIAA; 2000. Report No.: AIAA-2000-4565.
34. Enns D, Bugajski D, Hendrick, et al. Dynamic inversion: An evolving methodology for flight control design. *Int J Contr* 1994;**59**(1):71–91.
35. Smith P. A simplified approach to nonlinear dynamic inversion based flight control. Reston: AIAA; 1998. Report No.: AIAA-1998-4461.
36. Bachler J., Petit P. Incremental dynamic inversion flight control for the ACT/FHS EC 135 helicopter. *Proceedings of the vertical flight society 77th annual forum*. Fairfax: The Vertical Flight Society; 2021.
37. Kim CS, Ji CH, Koh GO, et al. Stability margin and structural coupling analysis of a hybrid INDI control for the fighter aircraft. *Int J Aeronaut Space Sci* 2021;**22**(5):1154–69.
38. Kim CS, Ji CH, Kim BS. Development of flight control law for improvement of uncommanded lateral motion of the fighter aircraft. *Int J Aeronaut Space Sci* 2020;**21**(4):1059–77.
39. Kim CS, Jin T, Koh GO, et al. Control law design to improve the unexpected pitch motion in slow down turn maneuver. *Proc Inst Mech Eng Part G J Aerosp Eng* 2022;**236**(1):123–39.
40. Kumtepe Y, Pollack T, Van Kampen EJ. Flight control law design using hybrid incremental nonlinear dynamic inversion. Reston: AIAA; 2022. Report No.: AIAA-2022-1597.
41. Bordignon K, Bessolo J. Control allocation for the X-35B. Reston: AIAA; 2002. Report No.: AIAA-2002-6020.
42. Harris JJ. F-35 flight control law design, development and verification. Reston: AIAA; 2018. Report No.: AIAA-2018-3516.
43. Steffensen R, Steinert A, Mbikayi Z, et al. Filter and sensor delay synchronization in incremental flight control laws. *Aerosp Syst* 2023;**6**(2):285–304.
44. Lyu HT, Ye Z, Chen YL, et al. Extended-state-observer-based angular acceleration estimation for supersonic aircraft lateral–directional control. *Appl Sci* 2023;**13**(11):6598.
45. Ludeña Cervantes TJ, Choi SH, Kim BS. Flight control design using incremental nonlinear dynamic inversion with fixed-lag smoothing estimation. *Int J Aeronaut Space Sci* 2020;**21**(4):1047–58.
46. Han JD, He YQ, Xu WL. Angular acceleration estimation and feedback control: An experimental investigation. *Mechatronics* 2007;**17**(9):524–32.
47. Liu YJ, Duan CY, Liu L, et al. Discrete-time incremental backstepping control with extended kalman filter for UAVs. *Electronics* 2023;**12**(14):3079.
48. Lin ZH, Liu QY, Liu SX, et al. Transition controller design of tilt-rotor UAV based on incremental nonlinear dynamic inversion. *IECON 2023–49th annual conference of the IEEE industrial electronics society*. Piscataway: IEEE Press; 2023.
49. Sieberling S, Chu QP, Mulder JA. Robust flight control using incremental nonlinear dynamic inversion and angular acceleration prediction. *J Guid Contr Dyn* 2010;**33**(6):1732–42.
50. Pfeifle O, Fichter W. Cascaded incremental nonlinear dynamic inversion for three-dimensional spline-tracking with wind compensation. *J Guid Contr Dyn* 2021;**44**(8):1559–71.
51. Taherinezhad M, Ramirez-Serrano A. An enhanced incremental nonlinear dynamic inversion control strategy for advanced unmanned aircraft systems. *Aerospace* 2023;**10**(10):843.
52. Zhou L, Yang JT, Strampe T, et al. Incremental nonlinear dynamic inversion based path-following control for a hybrid quad-plane unmanned aerial vehicle. *Int J Robust Nonlinear Contr* 2023;**33**(17):10304–27.
53. Zhang JN, Wang J, Zhang FB, et al. Modeling and incremental nonlinear dynamic inversion control for a highly redundant flight system. Reston: AIAA; 2019. Report No.: AIAA-2019-1922.
54. Henkenjohann M, Nolte U, Henke C, et al. Novel cascaded incremental nonlinear dynamic inversion controller approach for a tiltrotor VTOL. *2023 International conference on unmanned aircraft systems (ICUAS)*. Piscataway: IEEE Press; 2023.
55. Suiçmez EC, Kutay AT. Full envelope nonlinear flight controller design for a novel electric VTOL (eVTOL) air taxi. *Aeronaut J* 2024;**128**(1323):966–93.
56. Bacon BJ, Ostroff AJ, Joshi SM. Reconfigurable NDI controller using inertial sensor failure detection & isolation. *IEEE Trans Aerosp Electron Syst* 2001;**37**(4):1373–83.
57. Steffensen R, Steinert A, Smeur EJJ. Nonlinear dynamic inversion with actuator dynamics: an incremental control perspective. *J Guid Contr Dyn* 2022;**46**(4):709–17.
58. Milz D, May MS, Looye G. Dynamic inversion-based control concept for transformational tilt-wing eVTOLs. Reston: AIAA; 2024. Report No.: AIAA-2024-1290.
59. Surmann D, Myschik S. Gain design for an INDI-based flight control algorithm for a conceptual lift-to-cruise vehicle. Reston: AIAA; 2024. Report No.: AIAA-2024-1590.
60. Milz D, Looye G. Tilt-wing control design for a unified control concept. Reston: AIAA; 2022. Report No.: AIAA-2022-1084.
61. Isidori A. Nonlinear control systems. *Communications and control engineering series*. Berlin: Springer; 1995. p. 162.
62. Schildkamp R, Chang J, Sodja J, et al. Incremental nonlinear control for aeroelastic wing load alleviation and flutter suppression. *Actuators* 2023;**12**(7):280.
63. Delansnay G, Vande Wouwer A. Design of a reference governor in a zonotopic framework applied to a quadrotor under feedback linearization control strategies. *J Intell Rob Syst* 2023;**109**(1):7.
64. Durham W, Bordignon KA, Beck R. *Aircraft control allocation*. Hoboken: Wiley; 2016.p.76.
65. Oppenheimer MW, Doman DB. A method for compensation of interactions between second-order actuators and control

allocators. *2005 IEEE aerospace conference*. Piscataway: IEEE Press; 2005.
66. Li X, Jiang Y, Zhang JN, et al. A method to compensate interaction between actuator dynamics and control allocator under incremental nonlinear dynamic inversion controller. *IOP Conf Ser: Mater Sci Eng* 2018;**428**(1):012048.
67. Arun Kishore WC, Sen S, Ray G, et al. Dynamic control allocation for tracking time-varying control demand. *J Guid Contr Dyn* 2008;**31**(4):1150–7.
68. Raab SA, Zhang JN, Bhardwaj P, et al. Consideration of control effector dynamics and saturations in an extended INDI approach. Reston: AIAA; 2019. Report No.: AIAA-2019-3267.
69. Li MT, Li JP, Tang Y, et al. An extended INDI approach and application to pitch rate control laws design of an aircraft. Reston: AIAA; 2021. Report No.: AIAA-2021-3005.
70. Bhardwaj P, Raab SA, Holzapfel F. Higher order reference model for continuous dynamic inversion control. Reston: AIAA; 2021. Report No.: AIAA-2021-1130.
71. Pollack T, Looye G, Van der Linden F. Design and flight testing of flight control laws integrating incremental nonlinear dynamic inversion and servo current control. Reston: AIAA; 2019. Report No.: AIAA-2019-0130.
72. Rieger S. Rate based control of control effectors [Semester thesis]. Munich: Technical University of Munich 2020 [German].
73. Meyer-Brügel W, Silvestre FJ. Unified control of flight dynamics and actuators using local accelerations. *J Guid Contr Dyn* 2024;**47**(9):1920–38.
74. Acquatella BP, van Ekeren W, Chu QP. PI(D) tuning for flight control systems via incremental nonlinear dynamic inversion. *IFAC-PapersOnLine* 2017;**50**(1):8175–80.
75. Steffensen R, Steinert A, Holzapfel F. Incremental control as an enhanced and robust implementation of gain scheduled controllers avoiding hidden coupling terms. *Aerosp Sci Technol* 2023;**141**:108500.
76. Rugh WJ, Shamma JS. Research on gain scheduling. *Automatica* 2000;**36**(10):1401–25.
77. Kaminer I, Pascoal AM, Khargonekar PP, et al. A velocity algorithm for the implementation of gain-scheduled controllers. *Automatica* 1995;**31**(8):1185–91.
78. Hellmundt F, Wildschek A, Maier R, et al. Comparison of L1 adaptive augmentation strategies for a differential PI baseline controller on a longitudinal F16 aircraft model. *Advances in aerospace guidance, navigation and control*. Cham: Springer International Publishing; 2015. p. 99–118.
79. Oelker HC, Osterhuber R, Hanel M. Experiences with eurofighter handling qualities testing. Reston: AIAA; 2009. Report No.: AIAA-2009-6321.
80. Moritz N, Osterhuber R. Three-stage gradient-based optimization scheme in design of feedback gains within eurofighter primary control laws. Reston: AIAA; 2006. Report No.: AIAA-2006-6311.
81. Osterhuber R, Hanel M, Hammon R. Realization of the eurofighter 2000 primary lateral/directional flight control laws with differential PI-algorithm. Reston: AIAA; 2004. Report No.: AIAA-2004-4751.
82. Schatz SP, Gabrys AC, Gierszewski DM, et al. Inner loop command interface in a modular flight control architecture for trajectory flights of general aviation aircraft. *2018 5th International conference on control, decision and information technologies (CoDIT)*. Piscataway: IEEE Press; 2018.
83. Steinert A, Holzapfel F, Steffensen R, et al. Flight test based gain tuning using non-parametric frequency domain methods. Reston: AIAA; 2021. Report No.: AIAA-2021-1424.
84. Steinert A, Steffensen R, Gierszewski D, et al. Experimental results of flight test based gain tuning. Reston: AIAA; 2022. Report No.: AIAA-2022-2296.
85. Steffensen R, Steinert A, Holzapfel F. Longitudinal incremental reference model for fly-by-wire control law using incremental non-linear dynamic inversion. Reston: AIAA; 2022. Report No.: AIAA-2022-1230.
86. Zhang FB, Holzapfel F. Flight control using physical dynamic inversion. Reston: AIAA; 2015. Report No.: AIAA-2015-1758.
87. Peter F, Leitão M, Holzapfel F. Adaptive augmentation of a new baseline control architecture for tail-controlled missiles using a nonlinear reference model. Reston: AIAA; 2012. Report No.: AIAA-2012-5037.
88. Steffensen R, Ginnell K, Holzapfel F. Practical system identification and incremental control design for a subscale fixed-wing aircraft. *Actuators* 2024;**13**(4):130.
89. Pfeifle O, Fichter W. Time-optimal incremental nonlinear dynamic inversion through deadbeat control. Reston: AIAA; 2022. Report No.: AIAA-2022-1596.
90. Hovakimyan N, Lavretsky E, Sasane AJ. Dynamic inversion for nonaffine-in-control systems via time-scale separation. Part I. *Proceedings of the 2005, American control conference*. Piscataway: IEEE Press; 2005.
91. Lavretsky E, Hovakimyan N. Adaptive dynamic inversion for nonaffine-in-control uncertain systems via time-scale separation. *Part II. J Dyn Contr Syst* 2008;**14**(4):505–35.
92. Eberhardt RL, Ward DG. Indirect adaptive flight control system interactions. *Int J Robust Nonlinear Contr* 1999;**9**(14):1013–31.
93. Brinker J, Wise K. Reconfigurable flight control for a tailless advanced fighter aircraft. Reston: AIAA; 1998. Report No.: AIAA-1998-4107.
94. Calise A, Lee S, Sharma M. Direct adaptive reconfigurable control of a tailless fighter aircraft. Reston: AIAA; 1998. Report No.: AIAA-1998-4108.
95. Johnson E, Calise A, El-Shirbiny H, et al. Feedback linearization with neural network augmentation applied to X-33 attitude control. Reston: AIAA; 2000. Report No.: AIAA-2000-4157.
96. Kannan SK, Johnson EN. Adaptive control of systems in cascade with saturation. *49th IEEE conference on decision and control (CDC)*. Piscataway: IEEE Press; 2010.
97. Lakshmikanth GS, Padhi R, Watkins JM, et al. Adaptive flight-control design using neural-network-aided optimal nonlinear dynamic inversion. *J Aerosp Inf Syst* 2014;**11**(11):785–806.
98. Chowdhary G, Johnson E. Adaptive neural network flight control using both current and recorded data. Reston: AIAA; 2007. Report No.: AIAA-2007-6505.
99. Rysdyk R, Calise A, Chen R, et al. Nonlinear adaptive control of tiltrotor aircraft using neural networks. Reston: AIAA; 1997. Report No.: AIAA-1997-5613.
100. Achtelik M, Bierling T, Wang J, et al. Adaptive control of a quadcopter in the presence of large/complete parameter uncertainties. Reston: AIAA; 2011. Report No.: AIAA-2011-1485.
101. Geiser M, Xargay E, Hovakimyan N, et al. L1 adaptive augmented dynamic inversion controller for a high agility UAV. Reston: AIAA; 2011. Report No.: AIAA-2011-6457.
102. Mühlegg M, Niermeyer P, Falconi GP, et al. L1 fault tolerant adaptive control of a hexacopter with control degradation. *2015 IEEE conference on control applications (CCA)*. Piscataway: IEEE Press; 2015.
103. Yoon HJ, Cichella V, Hovakimyan N. Robust adaptive control allocation for an octocopter under actuator faults. Reston: AIAA; 2016. Report No.: AIAA-2016-0635.
104. Bhardwaj P, Akkinapalli VS, Zhang J, et al. Adaptive augmentation of incremental nonlinear dynamic inversion controller for an extended F-16 model. Reston: AIAA; 2019. Report No.: AIAA-2019-1923.

105. Casavola A, Garone E. Fault-tolerant adaptive control allocation schemes for overactuated systems. *Int J Robust Nonlinear Contr* 2010;**20**(17):1958–80.
106. Li HX, Myschik S, Holzapfel F. Null-space-excitation-based adaptive control for an overactuated hexacopter model. *J Guid Contr Dyn* 2023;**46**(3):483–98.
107. Falconi GP, Angelov J, Holzapfel F. Adaptive fault–tolerant position control of a hexacopter subject to an unknown motor failure. *Int J Appl Math Comput Sci* 2018;**28**(2):309–21.
108. Hafner SF, Hosseini B, Holzapfel F. Least squares based adaptive control allocation. *2023 IEEE conference on control technology and applications (CCTA)*. Piscataway: IEEE Press; 2023.
109. Hafner S, Ryals AD, Li HX, et al. Kalman filter based adaptive control allocation. *2024 32nd Mediterranean conference on control and automation (MED)*. Piscataway: IEEE Press; 2024.
110. Raab S, Steinert A, Hafner S, et al. Toward efficient calculation of inverses in control allocation algorithms for safety-critical applications. *J Guid Contr Dyn* 2024;**47**(11):2316–32.
111. Lu P, van Kampen EJ, de Visser C, et al. Aircraft fault-tolerant trajectory control using incremental nonlinear dynamic inversion. *Contr Eng Pract* 2016;**57**:126–41.
112. Smit B, Pollack T, Van Kampen EJ. Adaptive incremental nonlinear dynamic inversion flight control for consistent handling qualities. Reston: AIAA; 2022. Report No.: AIAA-2022-1394.
113. Liu ZC, Zhang YF, Liang JJ, et al. Application of the improved incremental nonlinear dynamic inversion in fixed-wing UAV fight tests. *J Aerosp Eng* 2023;**35**(6):04022091.
114. Pollack T, Looye G, Van der Linden F. Design and flight testing of flight control laws integrating incremental nonlinear dynamic inversion and servo current control. Reston: AIAA; 2019. Report No.: AIAA-2019-0130.
115. Steinleitner A, Frenzel V, Pfeifle O, et al. Automatic take-Off and landing of tailwheel aircraft with incremental nonlinear dynamic inversion. Reston: AIAA; 2022. Report No.: AIAA-2022-1228.
116. Simplício P, Pavel MD, van Kampen E, et al. An acceleration measurements-based approach for helicopter nonlinear flight control using incremental nonlinear dynamic inversion. *Contr Eng Pract* 2013;**21**(8):1065–77.
117. Pavel MD, Shanthakumaran P, Chu QP, et al. Incremental nonlinear dynamic inversion for the apache AH-64 helicopter control. *J Am Helicopter Soc* 2020;**65**(2):1–16.
118. Sman E, Smeur E, Remes B, et al. Incremental nonlinear dynamic inversion and multihole pressure probes for disturbance rejection control of fixed-wing micro air vehicles. *International micro air vehicle conference and flight competition (IMAV) 2017*. London: SAGE Publications Inc; 2017.
119. Smeur EJJ, de Croon GCHE, Chu Q. Cascaded incremental nonlinear dynamic inversion for MAV disturbance rejection. *Contr Eng Pract* 2018;**73**:79–90.
120. Di Francesco G, Mattei M. Modeling and incremental nonlinear dynamic inversion control of a novel unmanned tiltrotor. *J Aircr* 2015;**53**(1):73–86.
121. Bhardwaj P, Raab SA, Zhang JN, et al. Integrated reference model for a tilt-rotor vertical take-off and landing transition UAV. Reston: AIAA; 2018. Report No.: AIAA-2018-3479.
122. Panish L, Bacic M. A generalized full-envelope outer-loop feedback linearization control strategy for transition VTOL aircraft. Reston: AIAA; 2023. Report No.: AIAA-2023-4511.
123. Hartmann P, Meyer C, Moormann D. Unified velocity control and flight state transition of unmanned tilt-wing aircraft. *J Guid Contr Dyn* 2017;**40**(6):1348–59.
124. Tal E, Karaman S. Global incremental flight control for agile maneuvering of a tailsitter flying wing. *J Guid Contr Dyn* 2022;**45**(12):2332–49.
125. De Ponti TML, Smeur EJJ, Remes BWD. Incremental nonlinear dynamic inversion controller for a variable skew quad plane. *2023 international conference on unmanned aircraft systems (ICUAS)*. Piscataway: IEEE Press; 2023.
126. Myschik S, Kinast L, Huemer M, et al. Development of a flight control system for a cyclocopter UAV demonstrator. Reston: AIAA; 2022. Report No.: AIAA-2022-3282.
127. Azinheira J, Carvalho R, Paiva E, et al. Hexa-propeller airship for environmental surveillance and monitoring in Amazon rainforest. *Aerospace* 2024;**11**(4):249.
128. Steffensen R, Gabrys A, Holzapfel F. Flight envelope protections using phase plane limits and backstepping control. *5th CEAS conference on guidance, navigation and control*. Brussels: CEAS; 2019.
129. Gabrys AC, Steffensen R, de Angelis Cordeiro R, et al. Integration of phase plane flight envelope protections in cascaded incremental flight control. *IFAC-PapersOnLine* 2019;**52**(12):429–35.